

Supplementary Information

Constructing Photocatalytic Covalent Organic Frameworks with Aliphatic Linkers

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Section 1. Materials and general methods

1.1 Materials

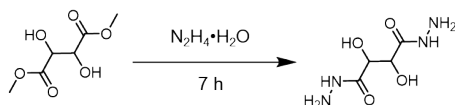
Solvents, reagents and chemicals were commercially available and used without further purification. Butanedihydrazide, 1,3,5-triformylphloroglucinol, hydroxyacetic acid hydrazide, and acethydrazide were purchased from Tokyo Chemical Industry Shanghai Co. Ltd. aniline and phenylenediamine were purchased from Sigama-Aldrich.

1.2 General methods

The solid-state ^{13}C Cross-Polarization with Total Suppression of Spinning Sidebands (^{13}C CP-TOSS) spectra was recorded on JEOL JNM ECZ600R (600 MHz) solid-state nuclear magnetic resonance (NMR) spectrometer. ^1H spectra were recorded in solution at 400 MHz, using a Bruker Advance 400 NMR spectrometer. The UV-Vis absorption spectra were obtained on a SHIMADZU 2600 spectrophotometer in the frequency range of 200-800 cm^{-1} . Powder X-ray diffraction (PXRD) patterns were recorded on a Bruker AXS D8 Advance powder diffractometer using Ni-filtered $\text{Cu } K_{\alpha}$ radiation with steps of 0.02° . High-resolution transmission electron microscopies (HRTEM) were performed on a Talos F200X. Field emission scanning electron microscopies (FE-SEM) was performed on a Helios G4 UC microscope operating at an accelerating voltage of 5.0 kV. N_2 adsorption-desorption isotherms were measured on an ASAP 2020 PLUS HD88 apparatus at 77 K and the surface areas were calculated by the Brunauere-Emmette-Teller (BET) method. The structures of COFs were simulated using a hexagonal unit cell in Material Studio 2017 package. The femtosecond pump-probe TAS measurements were performed using a regenerative amplified Ti: sapphire laser system (Coherent; 800 nm, 70 fs, 6 mJ/pulse and 1 kHz repetition rate) as the laser source and a Femto-100 spectrometer (Time-Tech LLC). The laser wavelength was changed by using a TOPAS Optical Parametric Amplifier (OPA). We used the super-continuum generation from a thin CaF_2 plate as the probe light. The delay between the pump and probe pulses was controlled by a motorized delay stage. The pump pulses were chopped by a synchronizhopper at 500 Hz and the absorbance change was calculated with two adjacent probe pulses (pump-blocked and pump-unblocked).

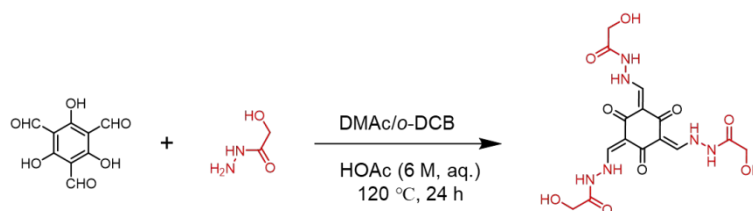
Section 2. Synthetic procedures

2.1 Synthesis of 2,3-dihydroxydicarbohydrazide



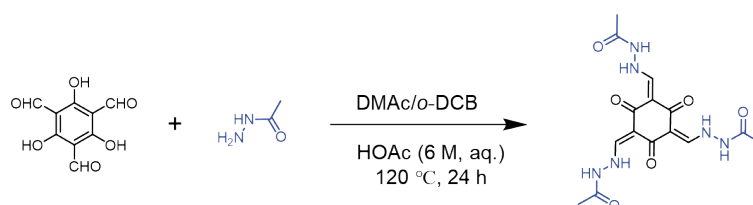
Dihydroxy-hydrazide was synthesized following a previously reported protocol^{S1}. Hydrazine monohydrate (3.1 mL, 116.8 mmol) was added to a stirred solution of diethyl L-tartrate (5 mL, 29.2 mmol) in ethanol (40 mL). The mixture was refluxed for 7 h. The precipitate was filtered off, washed with methanol, diethyl ether and dried to give the title compound as a white solid (4.61 g, yield 88%). ¹H NMR (400 MHz, DMSO-*d*₆, δ): 8.75 (*s*, 2H), 5.34 (*d*, 2H, $J = 7.5$ Hz), 4.21 (*t*, 6H, $J = 11.5$ Hz).

2.2 Synthesis of MC-TAH



To a 25 mL round bottom flask, 1,3,5-triformylphloroglucinol (Tp, 210 mg, 1 mmol), hydroxyacetic acid hydrazide (270 mg, 3 mmol) in a solution of 4.0 mL of Dimethylacetamide / 1.0 mL of 1,2-Dichlorobenzene / 1.0 mL of 6 M aqueous acetic acid. The mixture was stirred and heated to reflux. After 24 h, the reaction was cooled to room temperature, the precipitate was collected by filtration, washed with ethanol and THF, and dried under vacuum to give MC-TAH as a red solid (0.354 g, yield 83.1%). The model molecule was insoluble in conventional solvents (DCM, THF, DMF, DMSO), possibly attributed to the presence of strong hydrogen bonds. IR (powder, cm^{-1}): 1631 (C=O), 1591 (C=C), 1278 (C-N).

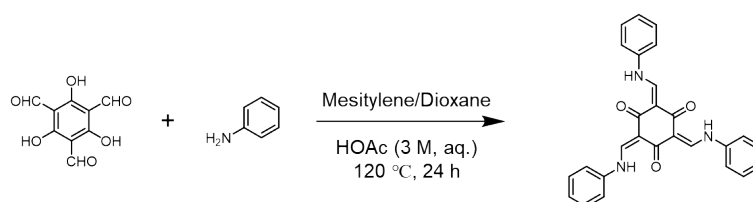
2.3 Synthesis of MC-BAH



To a 25 mL round bottom flask, 1,3,5-triformylphloroglucinol (Tp, 210 mg, 1 mmol), acethydrazide (222 mg, 3 mmol) in a solution of 2.5 mL of Dimethylacetamide / 2.5 mL of 1,2-Dichlorobenzene / 1.0 mL of 6 M aqueous acetic acid. The mixture was stirred and heated to reflux. After 24 h, the reaction was cooled to room temperature, precipitate was collected by filtration, washed with ethanol and THF, and dried under vacuum to give MC-BAH as an orange solid (0.301 g, yield 80.0%). The model molecule was insoluble in conventional solvents (DCM, THF, DMF,

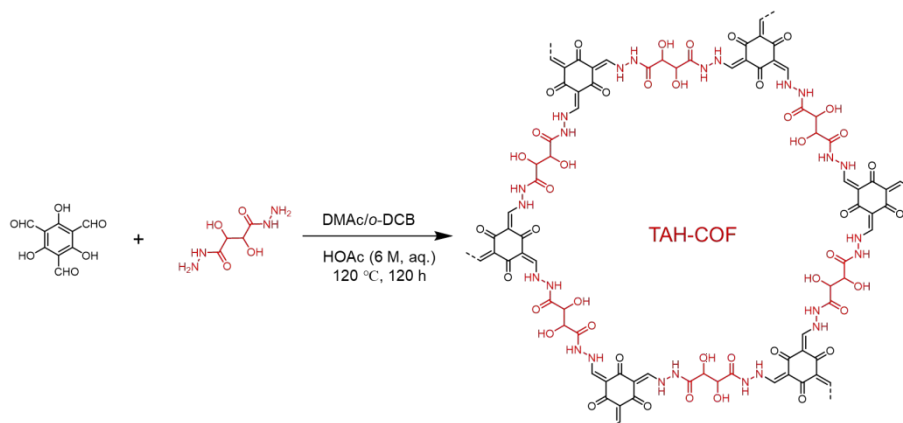
DMSO), possibly attributed to the presence of strong hydrogen bonds. IR (powder, cm^{-1}): 1683 (C=O), 1627 (C=C), 1230 (C-N).

2.4 Synthesis of MC-TPPA



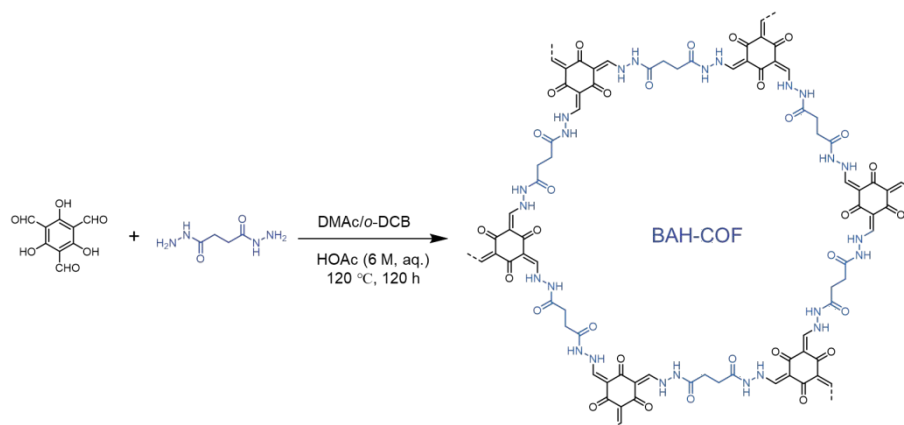
To a 25 mL round bottom flask, 1,3,5-triformylphloroglucinol (Tp, 210 mg, 1 mmol), aniline (280 mg, 3 mmol) in a solution of 5.0 mL of Mesitylene / 5.0 mL of Dioxane / 2.0 mL of 3 M aqueous acetic acid. The mixture was stirred and heated to reflux. After 24 h, the reaction was cooled to room temperature, precipitate was collected by filtration, washed with ethanol, and dried under vacuum to give MC-TPPA as a yellow solid (0.387 g, yield 88.3%). ^1H NMR (400 MHz, CDCl_3): 13.4 (d, 3H, $J = 12.9$ Hz), 8.80 (d, 3H, $J = 13.0$ Hz), 7.44-7.41 (m, 6H), 7.32-7.30 (m, 6H), 7.24-7.20 (m, 3H) ppm. IR (powder, cm^{-1}): 1616 (C=O), 1580 (C=C), 1283 (C-N).

2.5 Synthesis of TAH-COF



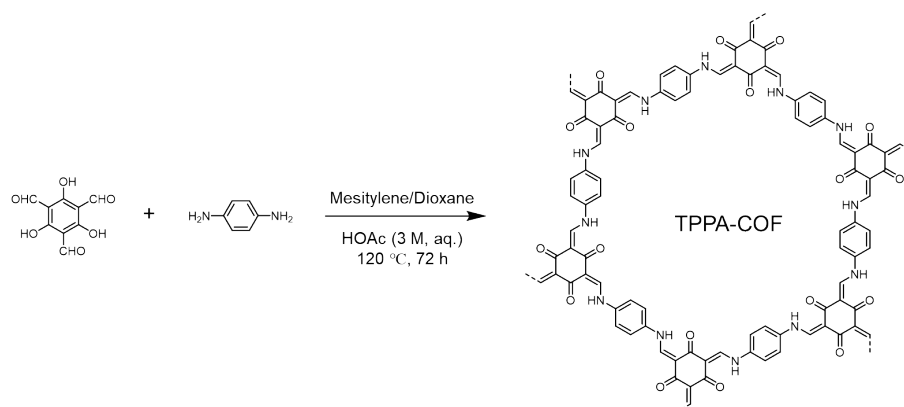
A Pyrex tube measuring o.d. $\times$ i.d. = 10 $\times$ 8 mm^2 was charged with 1,3,5-triformylphloroglucinol (Tp, 21 mg, 0.1 mmol) and 2,3-dihydroxydicarbohydrazide (TAH, 27 mg, 0.15 mmol) in a solution of 0.8 mL of Dimethylacetamide / 0.2 mL of 1,2-Dichlorobenzene / 0.2 mL of 6 M aqueous acetic acid. This mixture was sonicated for 10 min to form a homogeneous suspension. The tube was subsequently flash frozen at liquid N_2 temperature (77 K), degassed by three freeze-pump-thaw cycles, and then sealed. Finally, the reaction mixture was heated at 120 $^\circ\text{C}$ for 120 h under a static condition. After cooling to room temperature, the red precipitate was collected by filtration and washed with copious amounts of THF, DMF, H_2O and MeOH. The raw product was further carefully purified via Soxhlet extraction with THF and MeOH for 24 h, thus achieving the residual guest and oligomer removal. The material was dried under vacuum at 80 $^\circ\text{C}$ overnight to afford TAH-COF as a red powder (29 mg, yield 78%). IR (powder, cm^{-1}): 1628 (C=O), 1594 (C=C), 1281 (C-N).

2.6 Synthesis of BAH-COF



A Pyrex tube measuring o.d. $\times$ i.d. = 10 $\times$ 8 mm² was charged with 1,3,5-triformylphloroglucinol (Tp, 21 mg, 0.1 mmol) and Butanedihydrazide (BAH, 22 mg, 0.15 mmol) in a solution of 0.5 mL of Dimethylacetamide / 0.5 mL of 1,2-Dichlorobenzene / 0.2 mL of 6 M aqueous acetic acid. This mixture was sonicated for 10 min to form a homogeneous suspension. The tube was subsequently flash frozen at liquid N₂ temperature (77 K), degassed by three freeze-pump-thaw cycles, and then sealed. Finally, the reaction mixture was heated at 120 °C for 120 h under a static condition. After cooling to room temperature, the red precipitate was collected by filtration and washed with copious amounts of THF, DMF, H₂O and MeOH. The raw product was further carefully purified via Soxhlet extraction with THF and MeOH for 24 h, thus achieving the residual guest and oligomer removal. The material was dried under vacuum at 80 °C overnight to afford COF-BAD as a red powder (25 mg, yield 78%). IR (powder, cm⁻¹): 1630 (C=O), 1597 (C=C), 1281 (C-N).

2.7 Synthesis of TPPA-COF



TPPA-COF was synthesized following previously reported procedures^{S2}. A Pyrex tube measuring o.d. $\times$ i.d. = 10 $\times$ 8 mm² was charged with 1,3,5-triformylphloroglucinol (Tp, 63 mg, 0.3 mmol) and *p*-phenylenediamine (Pa, 48 mg, 0.45 mmol) in a solution of 1.5 mL of Mesitylene / 1.5 mL of Dioxane / 0.5 mL of 3 M aqueous acetic acid. This mixture was sonicated for 10 min to form a homogeneous suspension. The tube was subsequently flash frozen at liquid N₂ temperature (77 K), degassed by three freeze-pump-thaw cycles, and then flame sealed. Finally, the reaction mixture was heated at 120 °C for 72 h under a static condition. After cooling to room temperature, the red

precipitate was collected by filtration and washed with copious amounts of THF, DMF, H₂O and MeOH. The raw product was further carefully purified via Soxhlet extraction with THF and MeOH for 24 h, thus achieving the residual guest and oligomer removal. The material was dried under vacuum at 80 °C overnight to afford TPPA-COF as a red powder (81 mg, yield 85%). IR (powder, cm⁻¹): 1630 (C=O), 1594 (C=C), 1276 (C-N).

Section 3. Characterization of TAH-COF, BAH-COF, and TPPA-COF

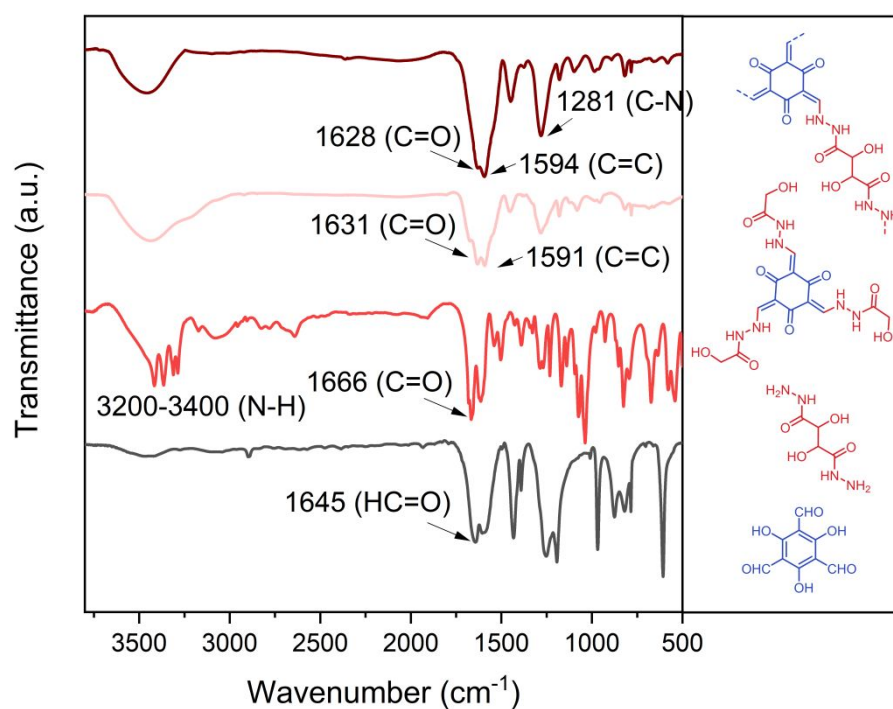


Figure S1. Comparison of FT-IR spectra of TAH-COF, model compound MC-TAH and corresponding starting monomers (TAH and TP).

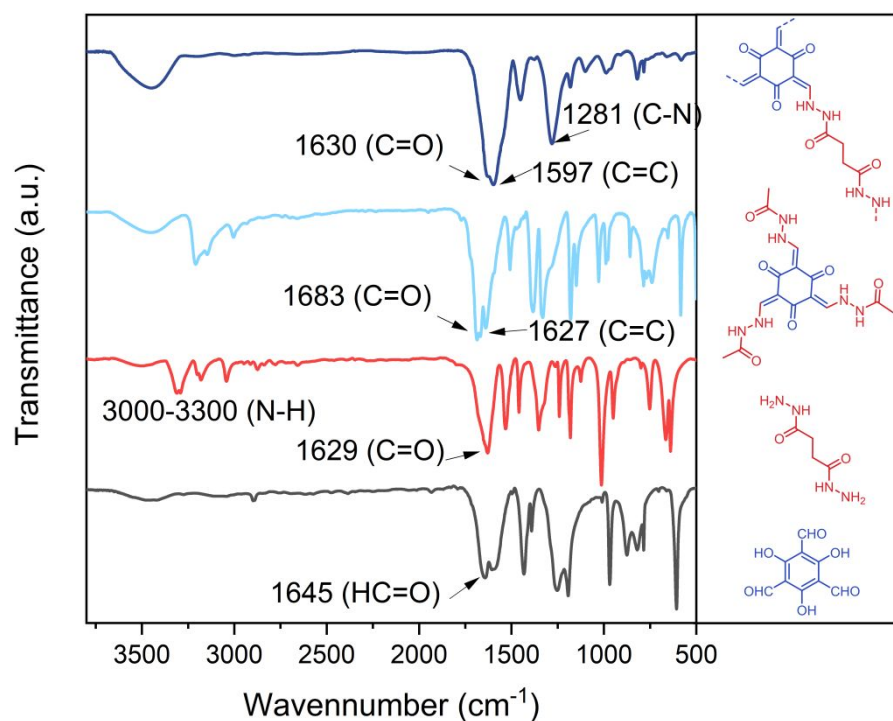


Figure S2. Comparison of FT-IR spectra of BAH-COF, model compound MC-BAH and corresponding starting monomers (BAH and TP).

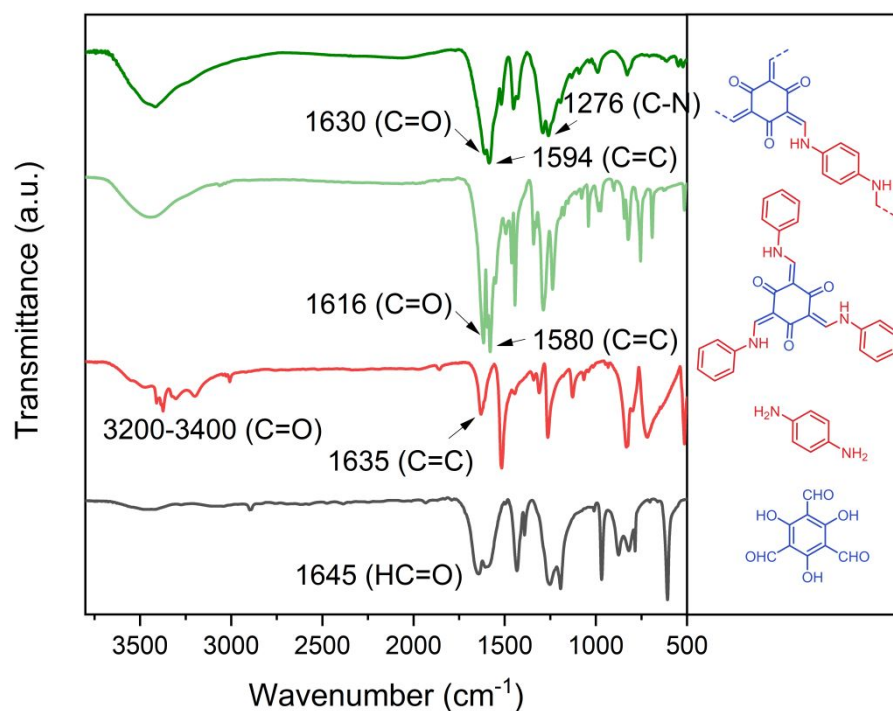


Figure S3. Comparison of FT-IR spectra of TPPA-COF, model compound MC-TPPA and corresponding starting monomers (PA and TP).

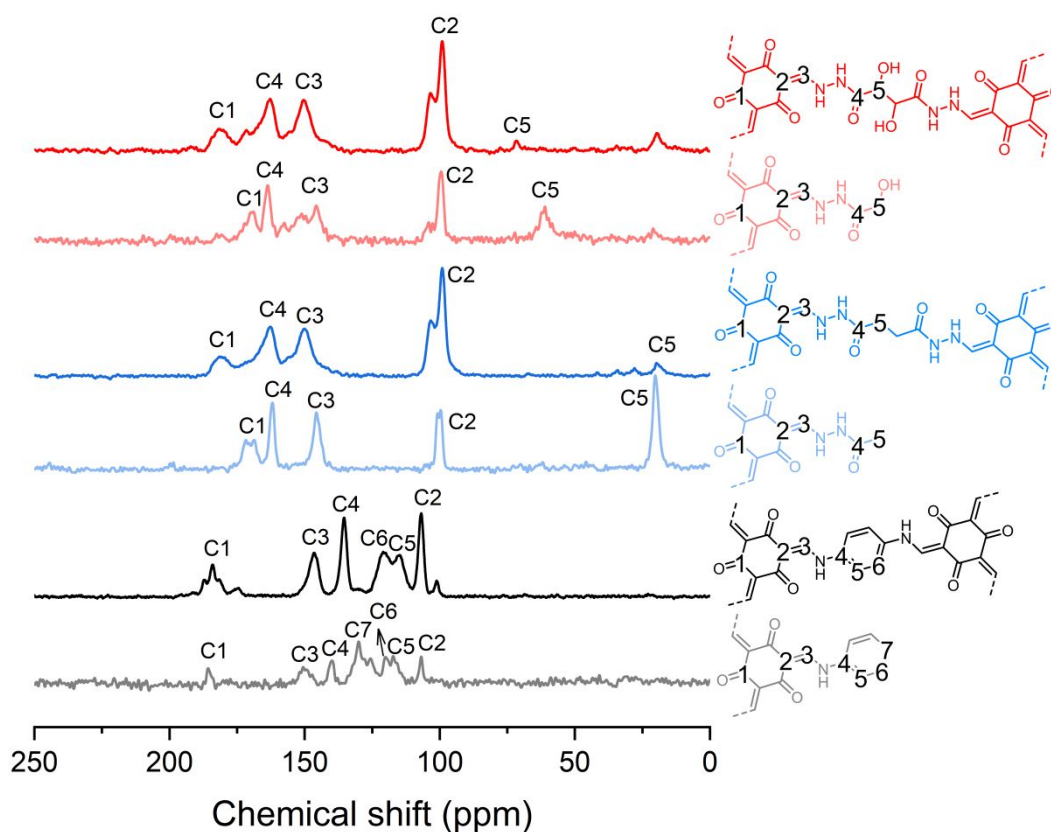


Figure S4. Solid-state ^{13}C CP-MAS NMR spectra of TAH-COF, BAH-COF, TPPA-COF, and corresponding model compounds.

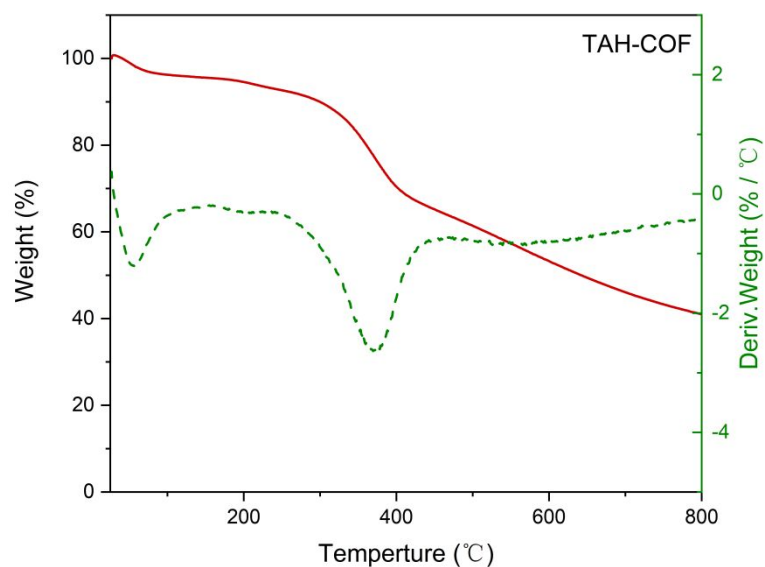


Figure S5. Thermogravimetric curve and corresponding weight loss for TAH-COF under nitrogen atmosphere. The mass loss observed above 100 °C could be attributed to the presence of water molecules adsorbed on the surface of the COF.

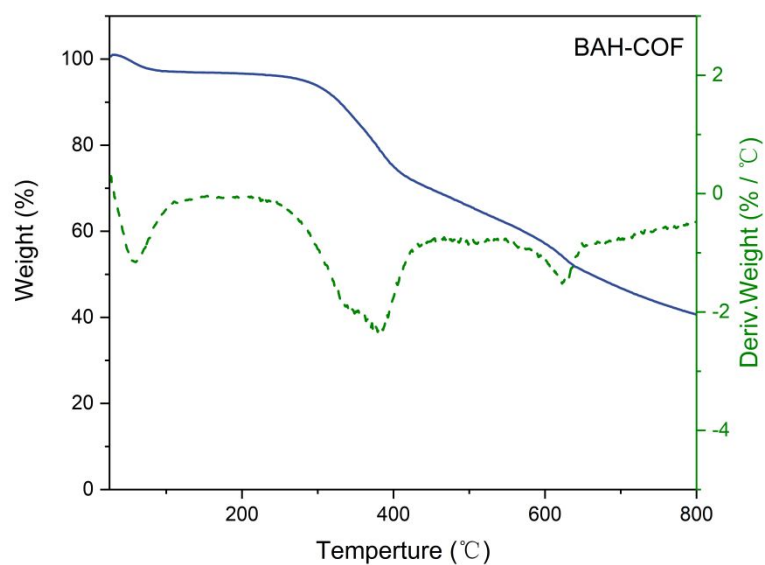


Figure S6. Thermogravimetric curve and corresponding weight loss for BAH-COF under nitrogen atmosphere. The mass loss observed above 100 °C could be attributed to the presence of water molecules adsorbed on the surface of the COF.

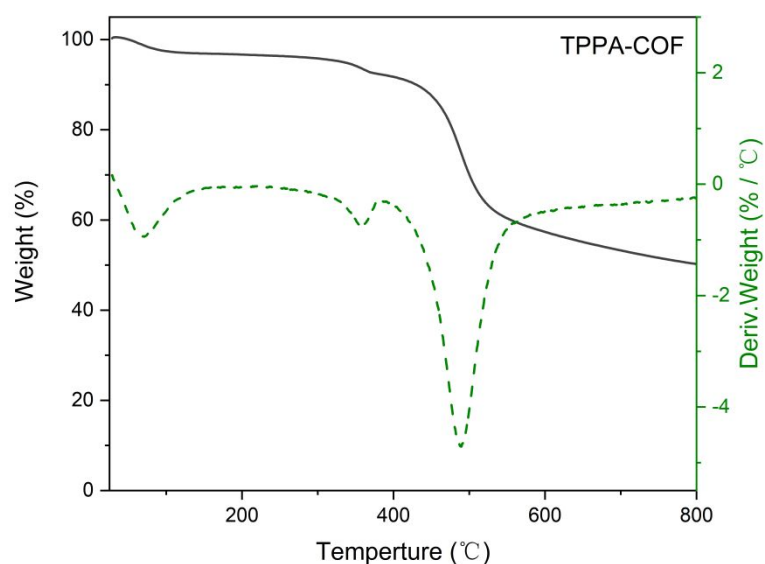


Figure S7. Thermogravimetric curve and corresponding weight loss for TPPA-COF under nitrogen atmosphere. The mass loss observed above 100 °C could be attributed to the presence of water molecules adsorbed on the surface of the COF.

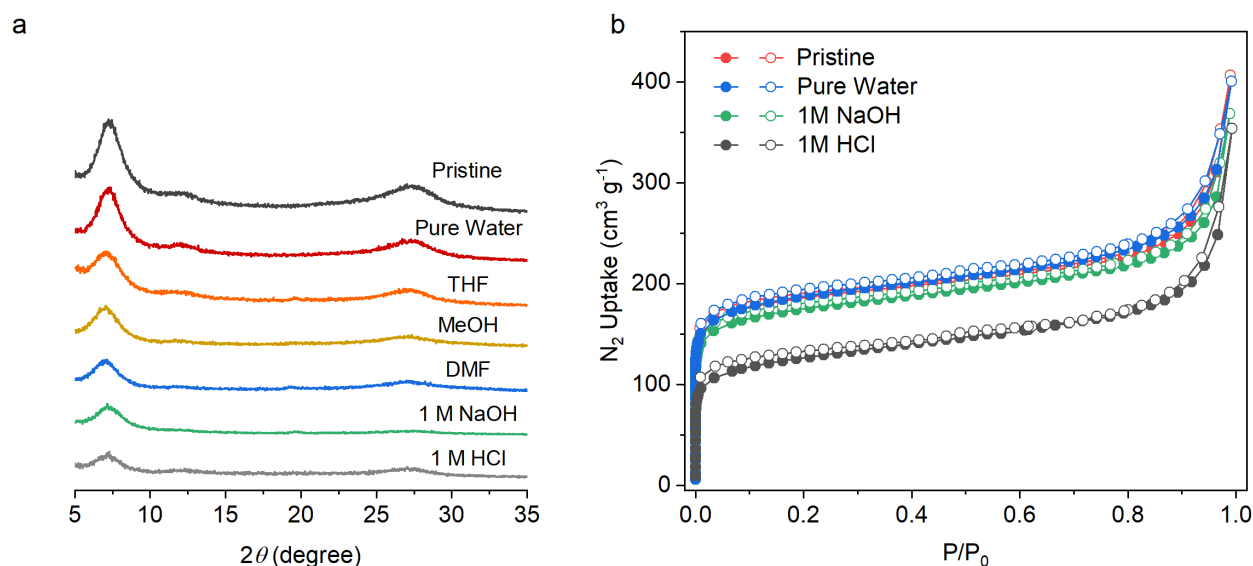


Figure S8. (a) PXRD patterns and (b) N₂ adsorption-desorption isotherms for TAH-COF after various chemical treatments for 24 h. The calculated BET surface areas are 703, 704, 657, 466 m² g⁻¹ for TAH-COF of pristine, pure water, 1 M NaOH and 1 M HCl, respectively. TAH-COF retained its crystalline structure after treatment under these conditions.

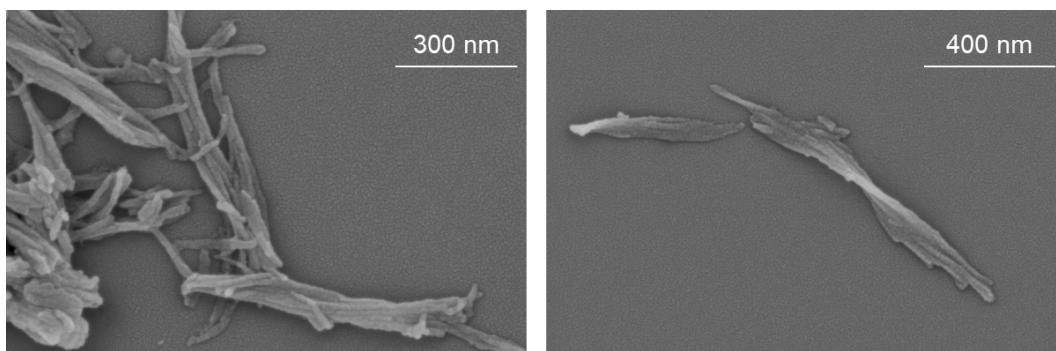


Figure S9. FE-SEM images of TAH-COF.

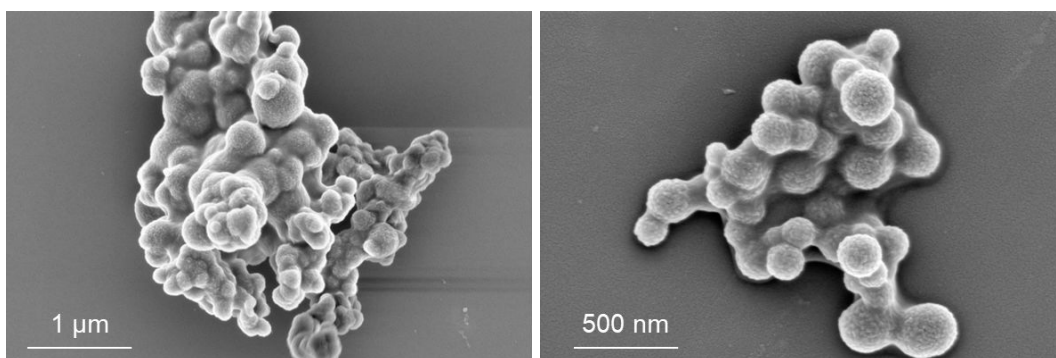


Figure S10. FE-SEM images of BAH-COF.

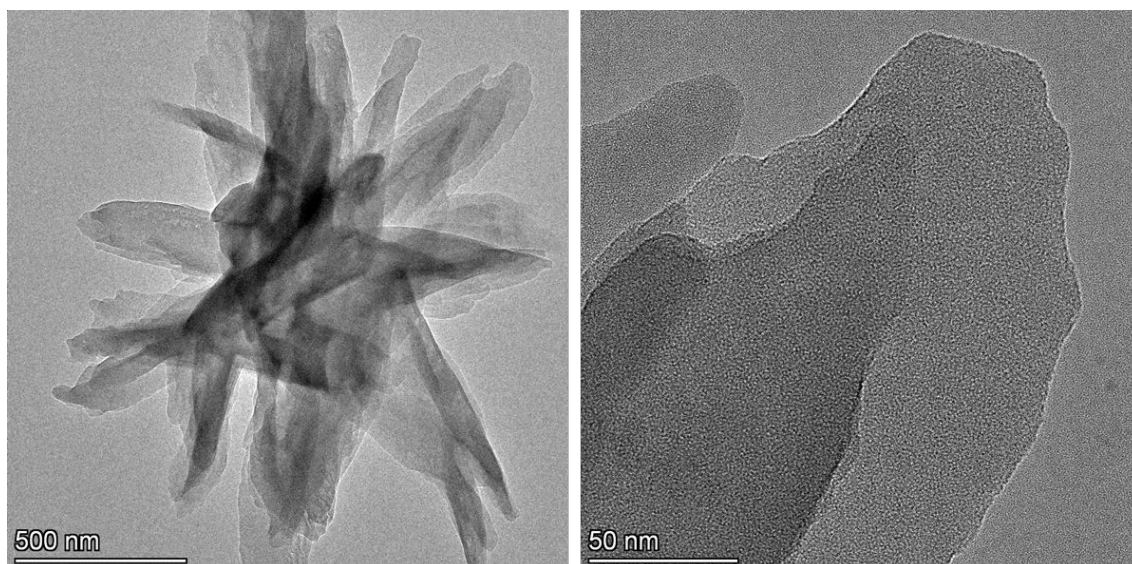


Figure S11. HRTEM images of TAH-COF.

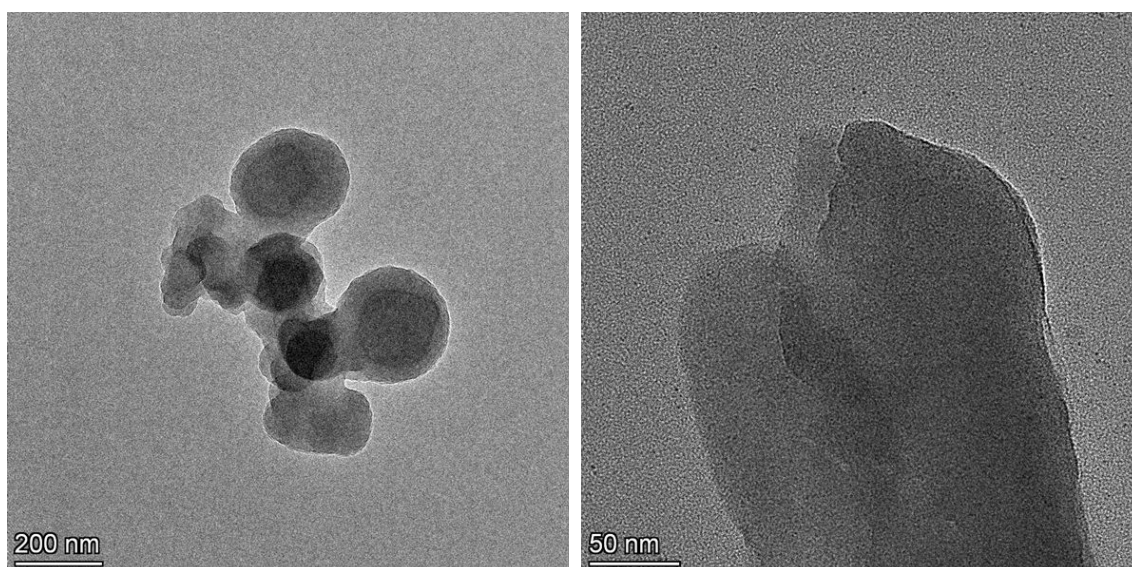


Figure S12. HRTEM images of BAH-COF.

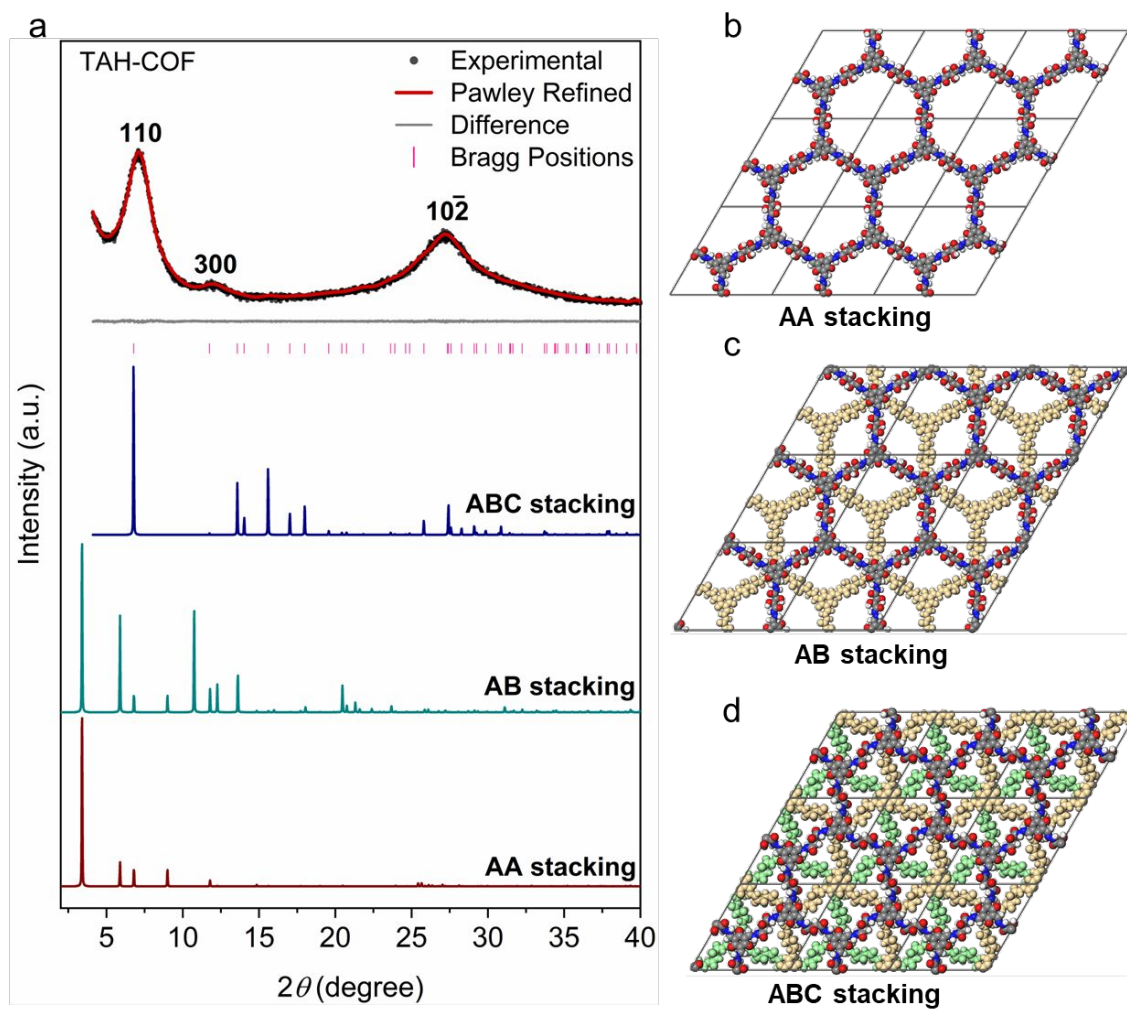


Figure S13. (a) Experimental, pawley refined, and simulated PXRD profiles of the TAH-COF with difference ($R_{wp} = 3.23\%$ and $R_p = 2.54\%$). The top view of eclipsed AA stacking (b), staggered AB stacking (c) and staggered ABC stacking (d) models for TAH-COF.

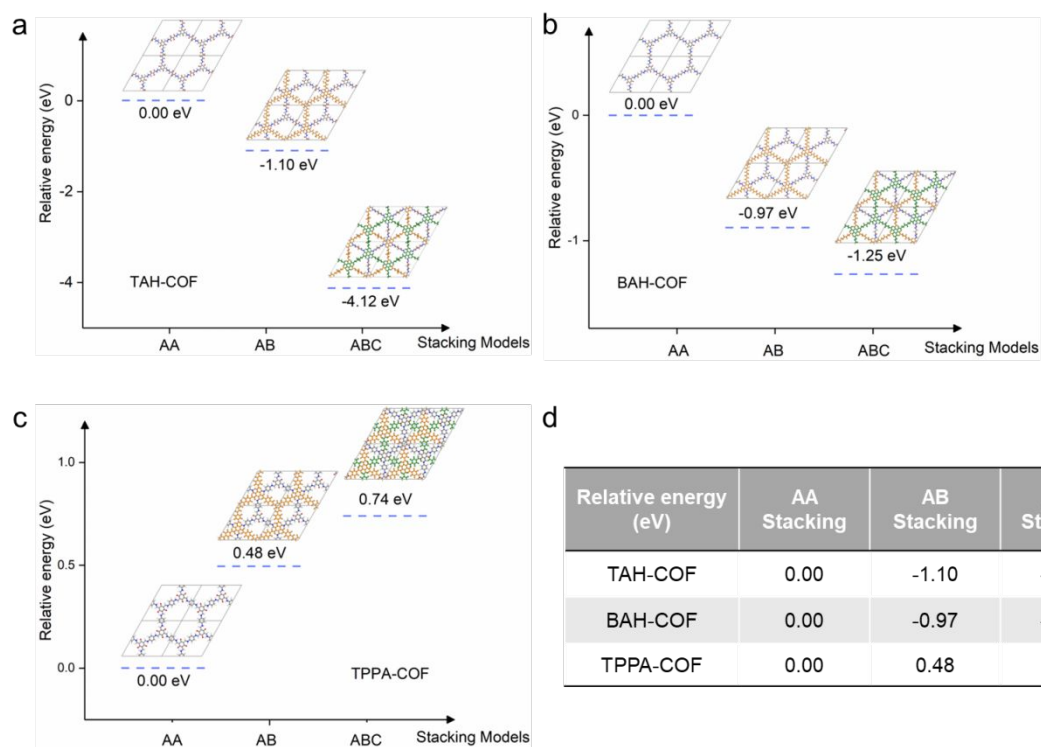


Figure S14. The relative crystal stacking energies of AA, AB, and ABC stacking models for (a) TAH-COF, (b) BAH-COF and (c) TPPA-COF. For TAH-COF and BAH-COF, the stacking energies for ABC are relatively lower than those for AA and AB. Conversely, in TAPA-COF, adopting AA stacking results in lower stacking energies.

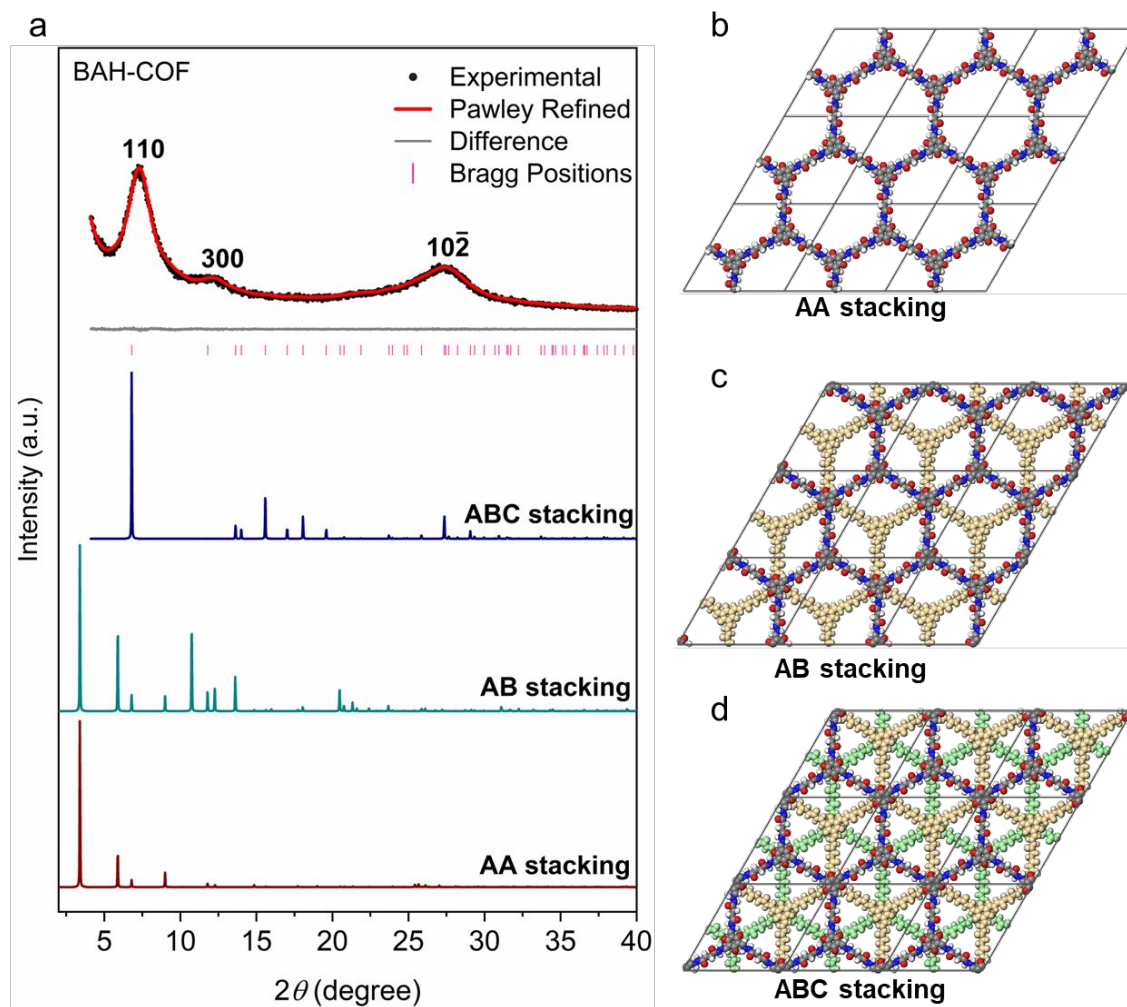


Figure S15. (a) Experimental, pawley refined, and simulated PXRD profiles of the BAH-COF with difference ($R_{wp} = 4.08\%$ and $R_p = 2.99\%$). The top view of eclipsed AA stacking (b), staggered AB stacking (c) and staggered ABC stacking (d) models for BAH-COF.

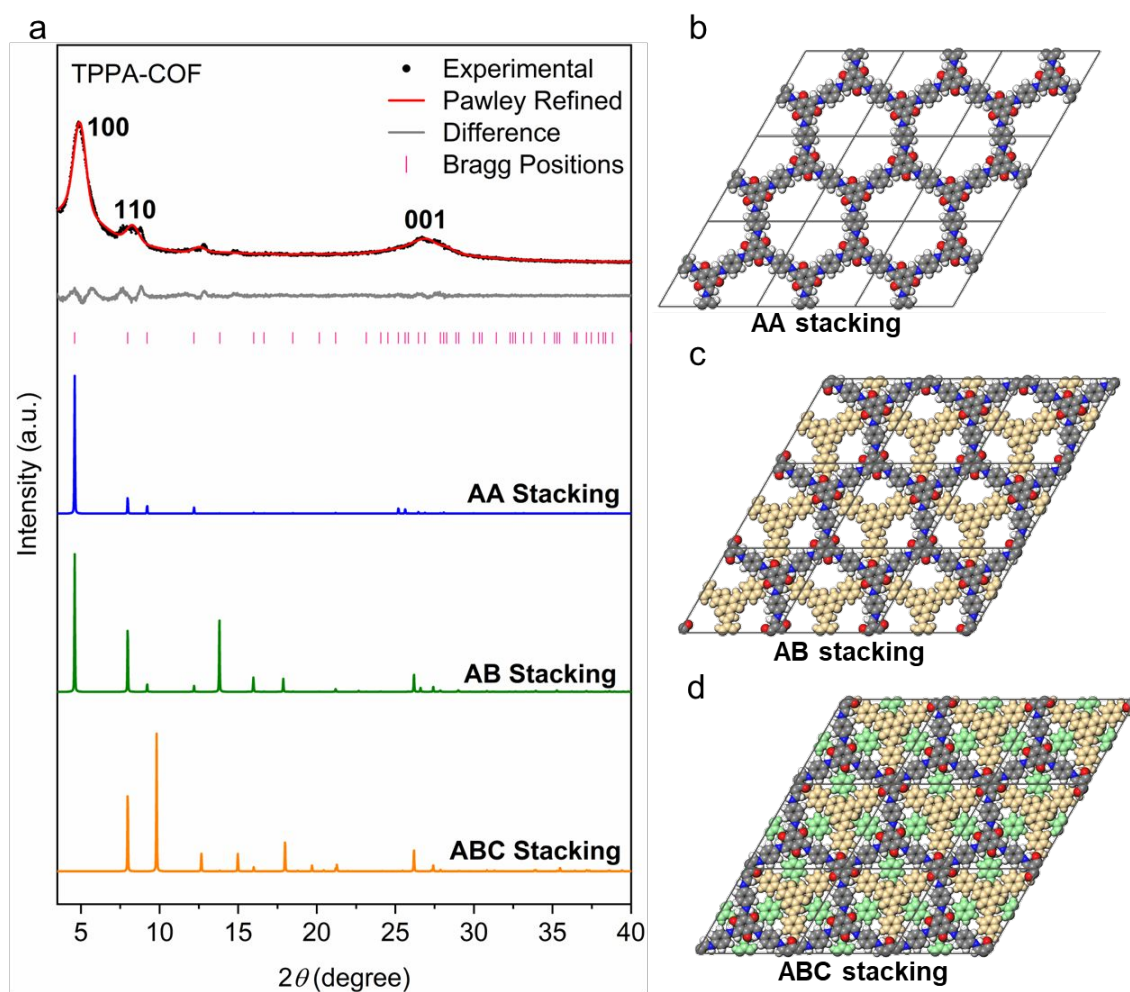


Figure S16. (a) Experimental, pawley refined, and simulated PXRD profiles of the TPPA-COF with difference ($R_{wp} = 6.00\%$ and $R_p = 4.54\%$). The top view of eclipsed AA stacking (b), staggered AB stacking (c) and staggered ABC stacking (d) models for TPPA-COF.

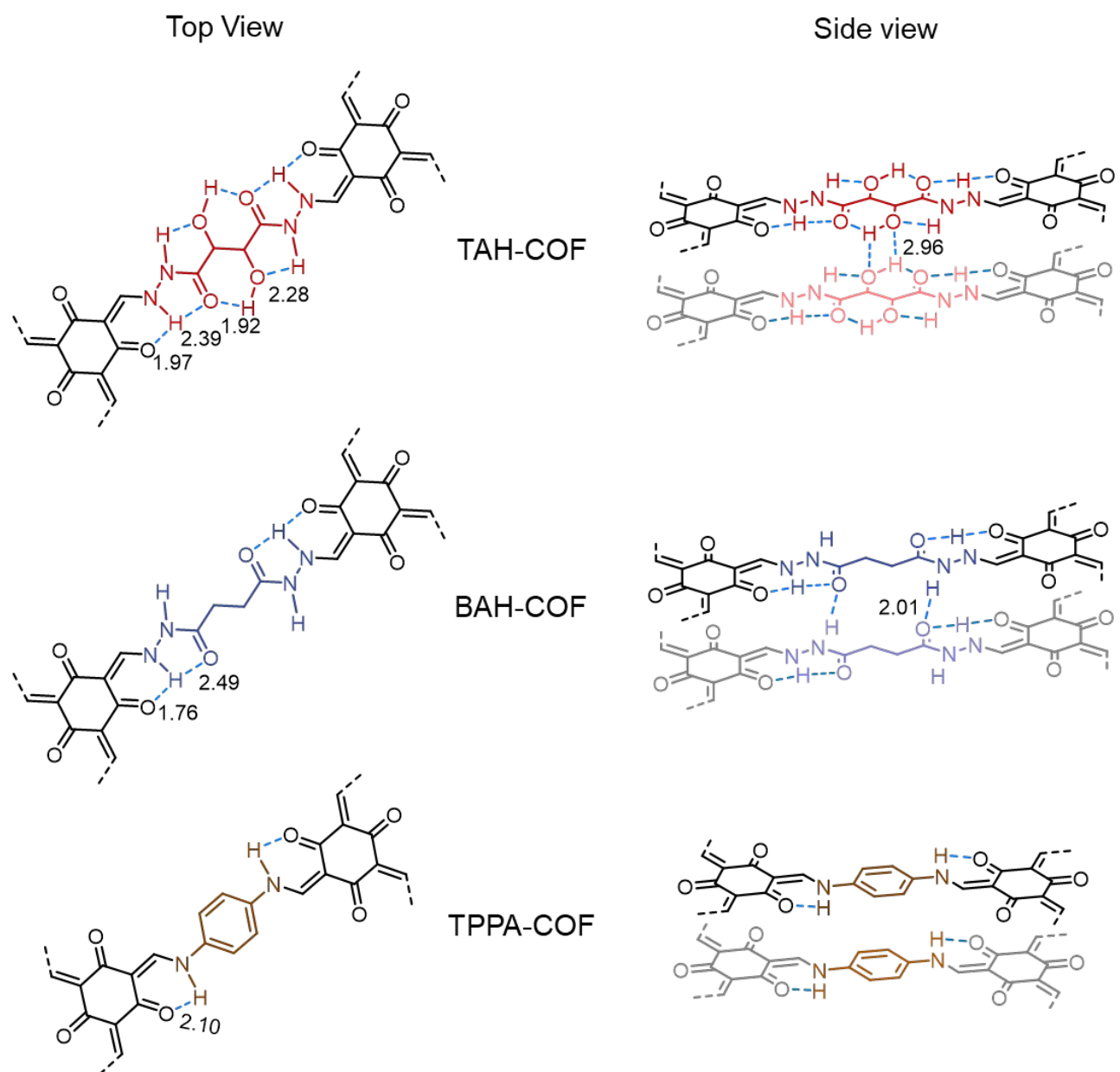


Figure S17. The different hydrogen bonding networks distribution for TAH-COF, BAH-COF, and TPPA-COF derived from structural models.

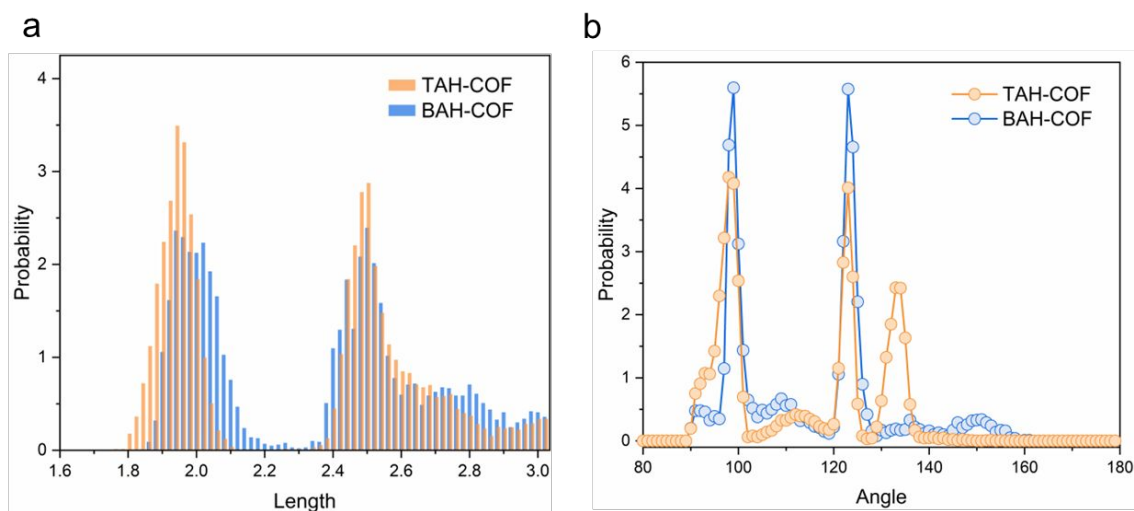


Figure S18. The statistical distributions of hydrogen bonding (a) length, and (b) angle per unit cell of TAH-COF and BAH-COF derived from structural models.

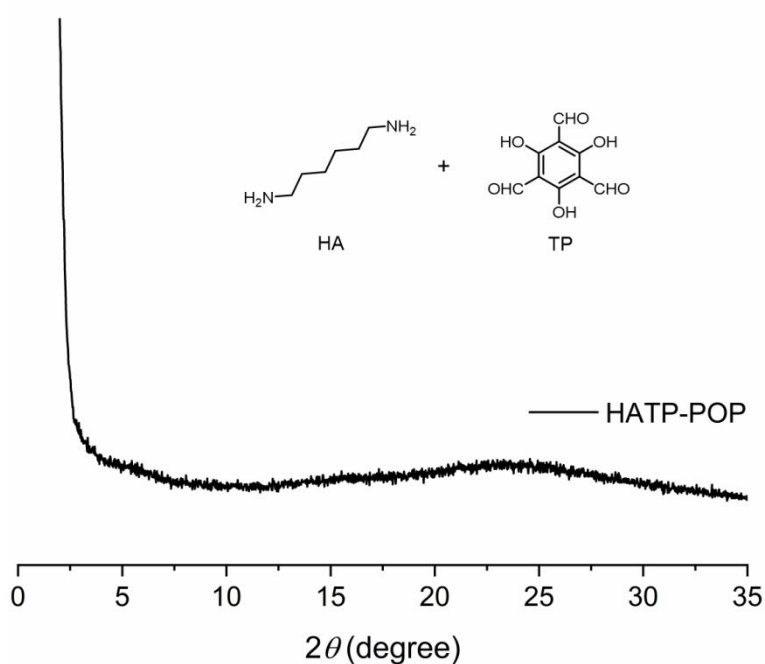


Figure S19. The PXRD measurement for HATP-POP. Synthesis of aliphatic linker-based COF using HA ligand without proton donor or acceptor resulted in the formation of amorphous polymers. This highlights the significance of suitable proton donors and acceptors in promoting the crystallization of aliphatic linker-based COFs.

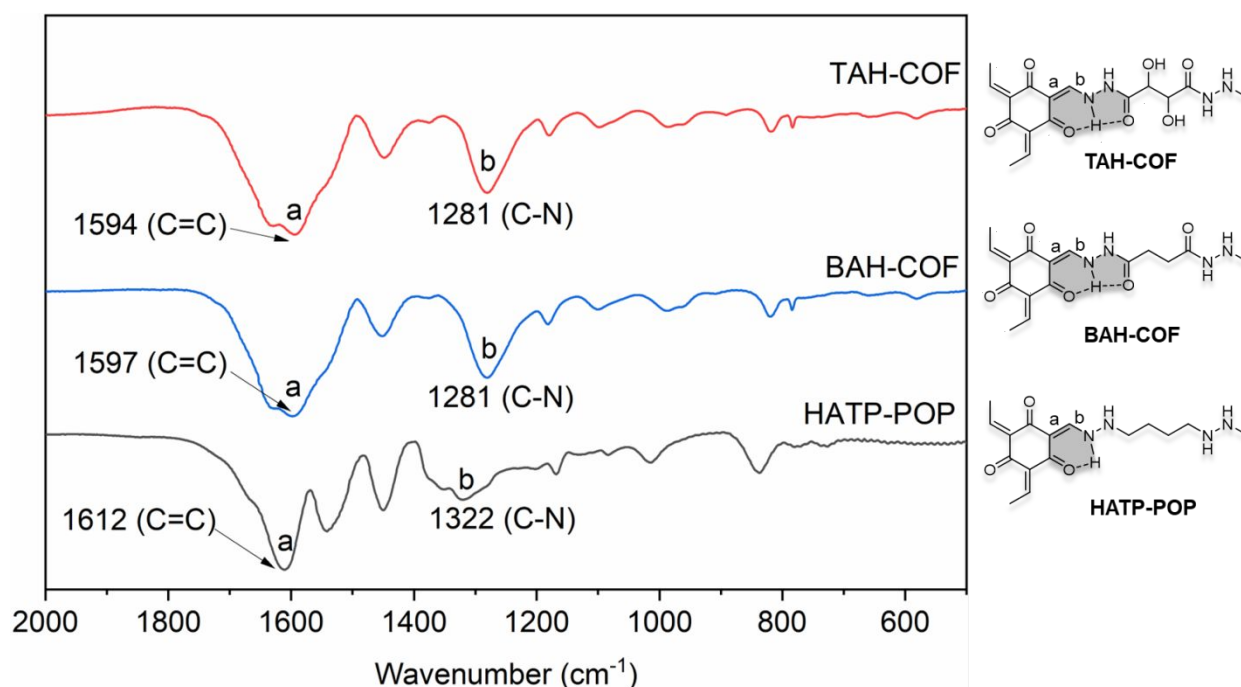


Figure S20. The FT-IR spectra of TAH-COF, BAH-COF, and HATP-POP. The infrared absorption frequencies of the C=C (a) and C-N bonds (b) in TAH-COF and BAH-COF are lowered due to the presence of hydrogen bonds, compared to HATP-POP.

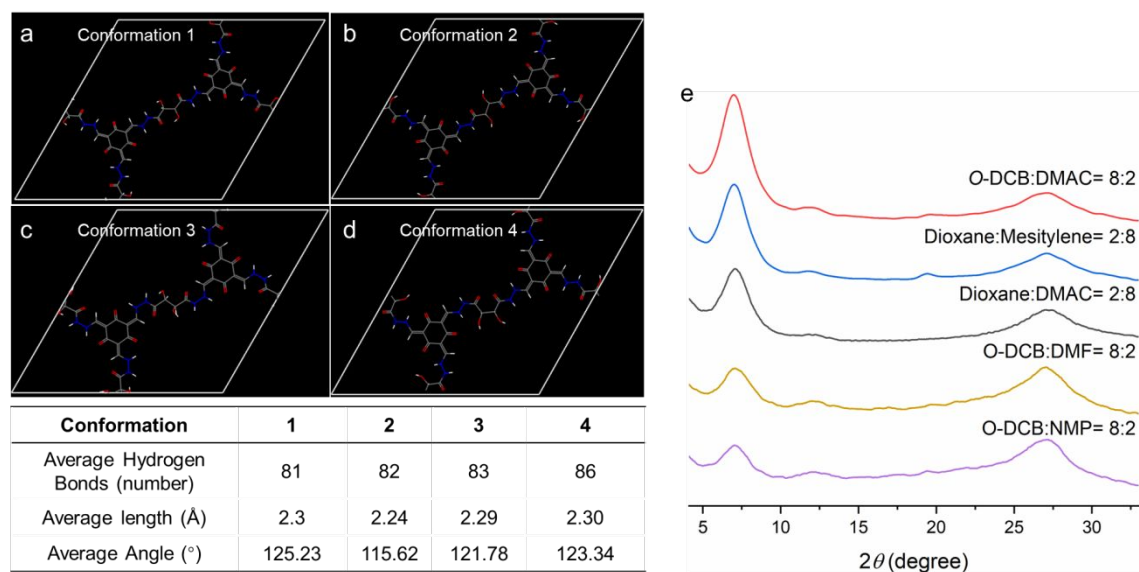


Figure S21. (a)-(d) TAH-COF structures based on various conformations simulated in Materials Studio. (e) PXRD of TAH-COF on different solvothermal reactions. These conformations (1-4) exhibit different types of hydrogen bonding network to stabilize the framework. However, Crystallization of TAH-COF using different solvent mixtures gave rise to very similar PXRD patterns. This indicates that the conformation 1 we simulated is likely a thermodynamically stable product.

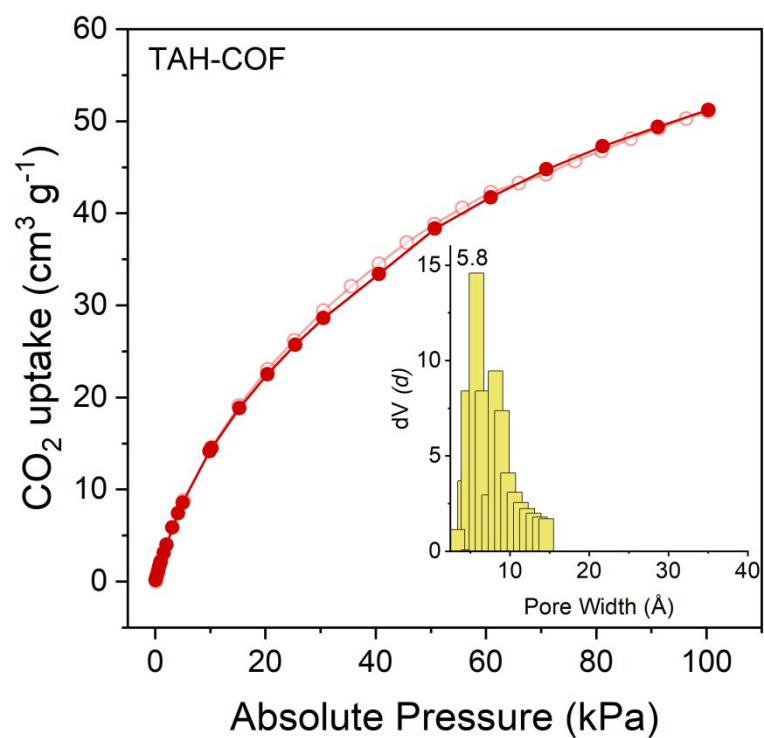


Figure S22. CO₂ adsorption-desorption isotherms were measured at 273 K for TAH-COF. The calculated pore size distributions showed a peak centered around 5.8 Å, which is in good agreement with the value obtained from the N₂ adsorption-desorption isotherms measured at 77 K.

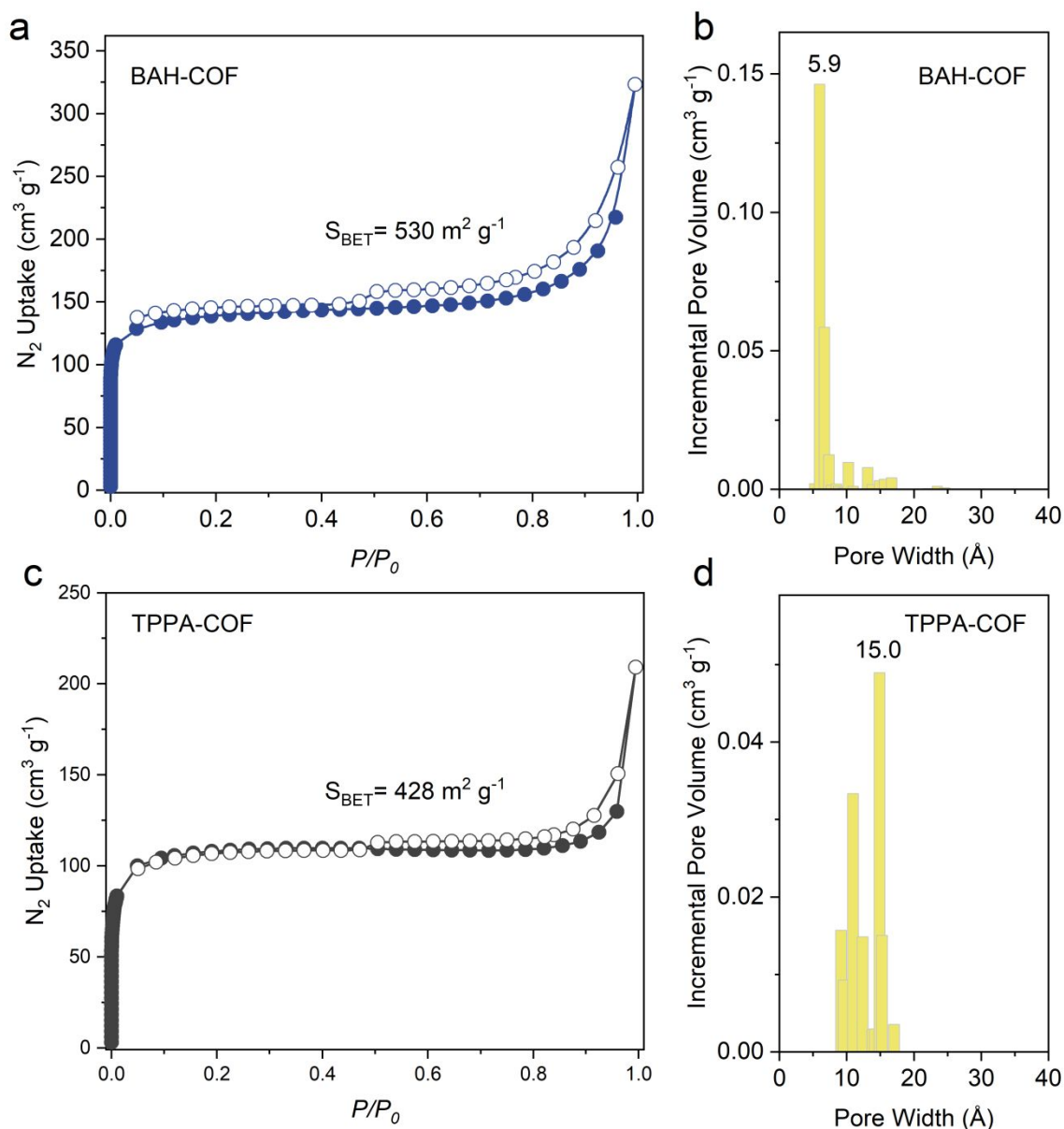


Figure S23. N_2 adsorption-desorption isotherms were measured at 77 K for BAH-COF and TPPA-COF. The narrow pore size distributions, obtained by fitting the adsorption isotherms using nonlocal density functional theory, were found to be in precise agreement with their proposed structural models.

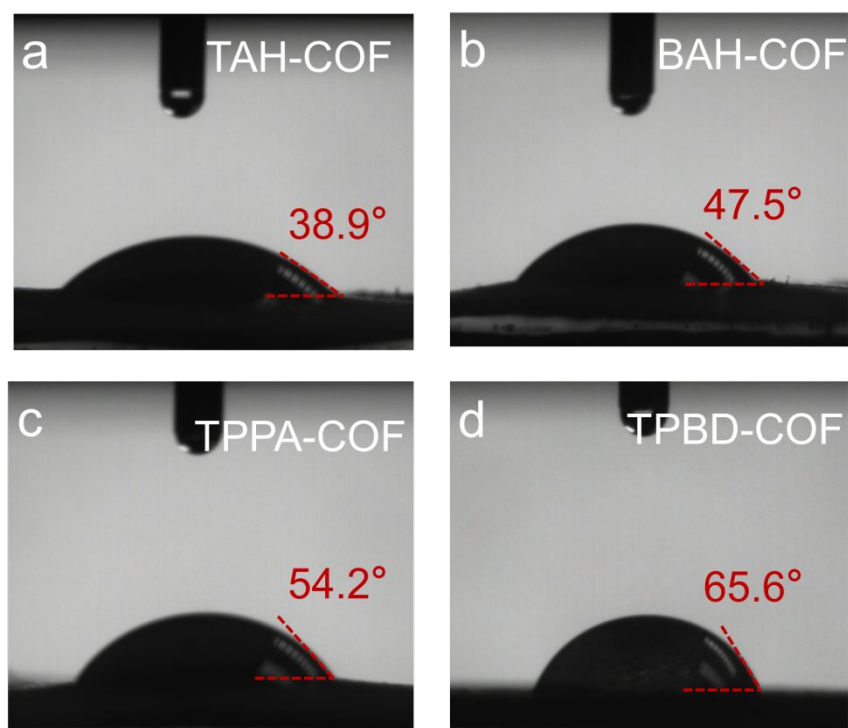


Figure S24. Water contact angle tests of TAH-COF, BAH-COF, TPPA-COF, and TPBD-COF.

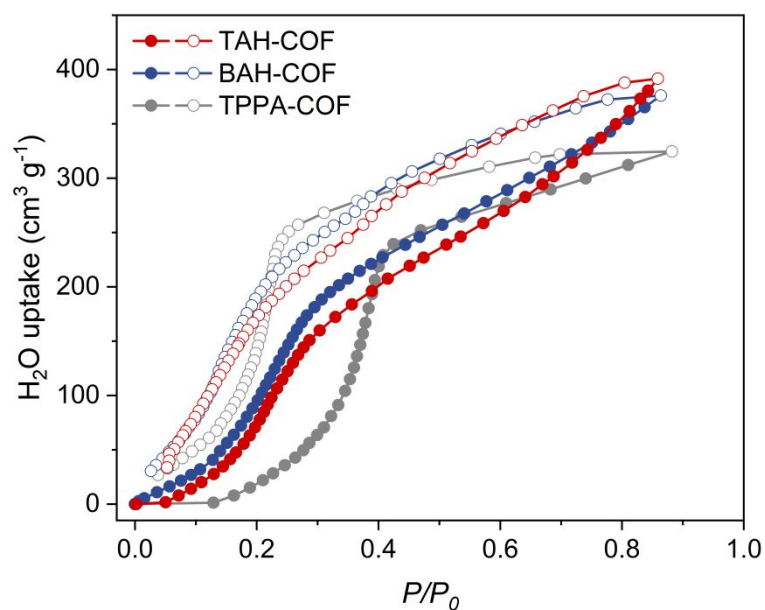


Figure S25. Water sorption curves of TAH-COF, BAH-COF, and TPPA-COF at 298 K. The water uptake capacity was significantly higher for TAH-COF ($77.8 \text{ cm}^3 \text{g}^{-1}$) and BAH-COF ($86.2 \text{ cm}^3 \text{g}^{-1}$) at low partial pressure ($P/P_0 < 0.2$) compared to TPPA-COF ($21.2 \text{ cm}^3 \text{g}^{-1}$).

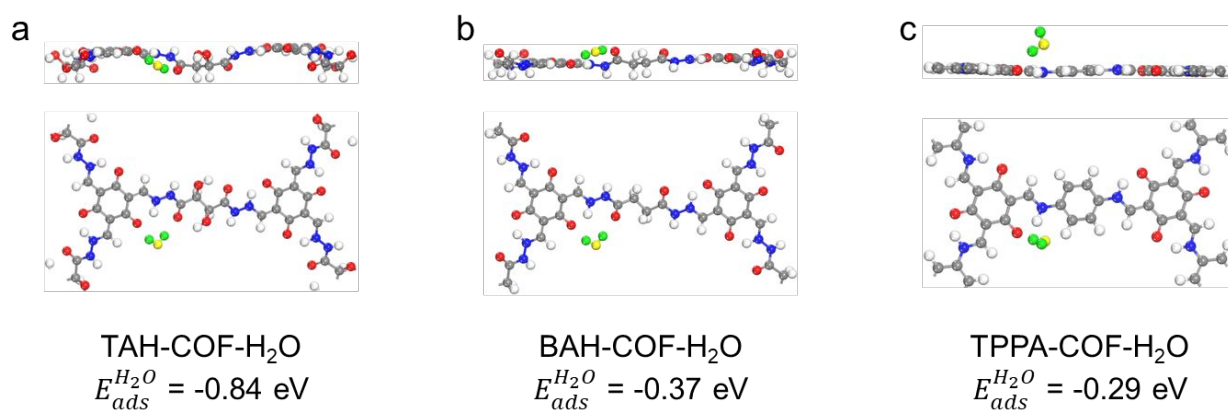


Figure S26. The binding energies between the segments of (a) TAH-COF, (b) BAH-COF and (c) TPPA-COF and water molecules. The TAH-COF with more polar groups exhibits significantly stronger interactions compared to the BAH-COF and TPPA-COF.

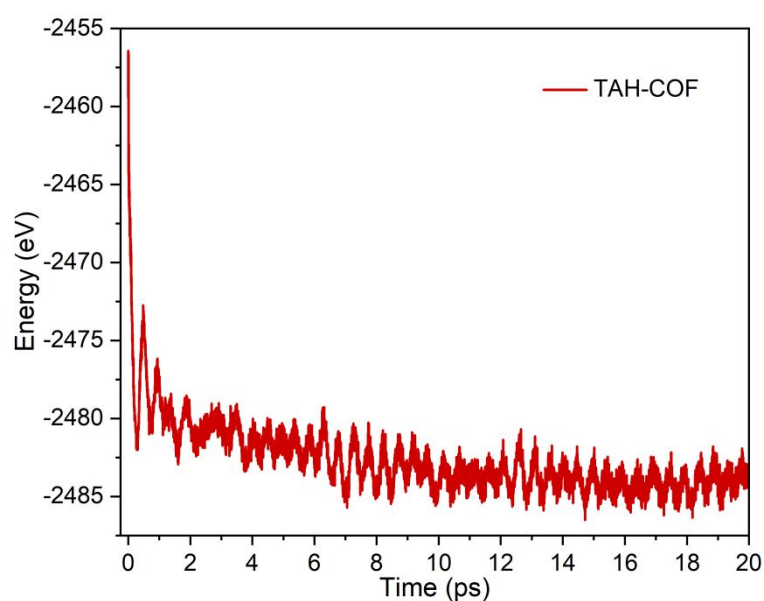


Figure S27. The mean squared displacement curves of molecular dynamics (MD) simulations for the water adsorption of TAH-COF, suggesting MD results were equilibrated for the calculated time.

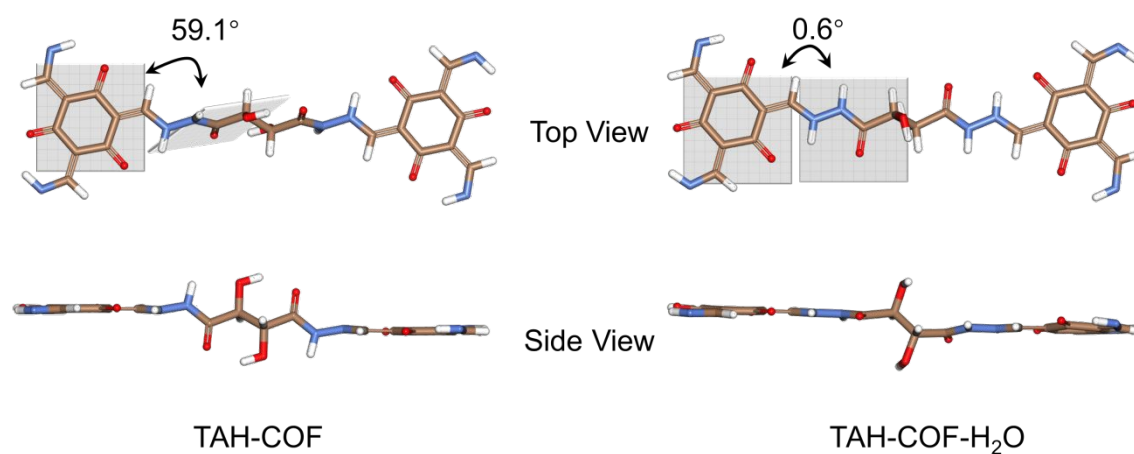


Figure S28. MD simulation snapshots of TAH-COF before and after water adsorption equilibrium (C, brown; N, blue; O, red; H, white).

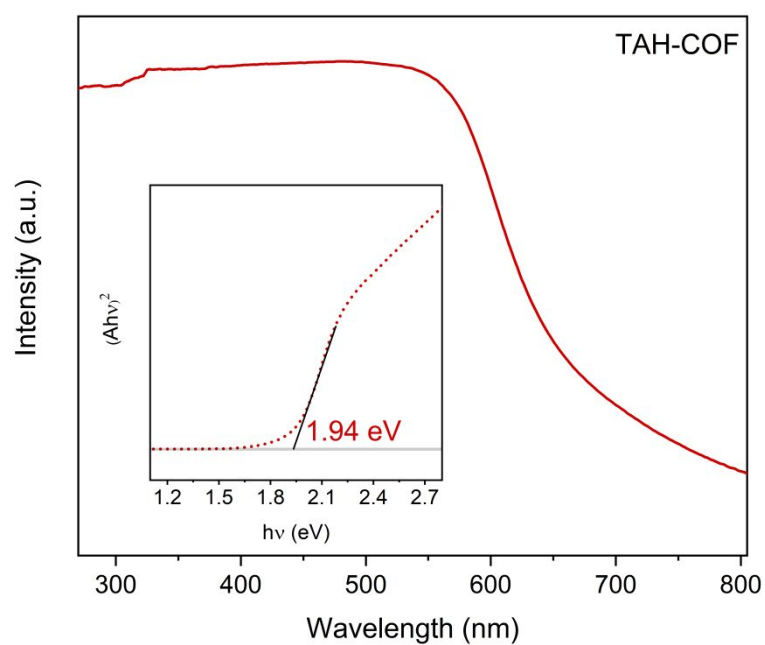


Figure S29. UV/vis diffuse reflectance spectra (DRS) spectra of TAH-COF. The insert figure is the corresponding Tauc plot.

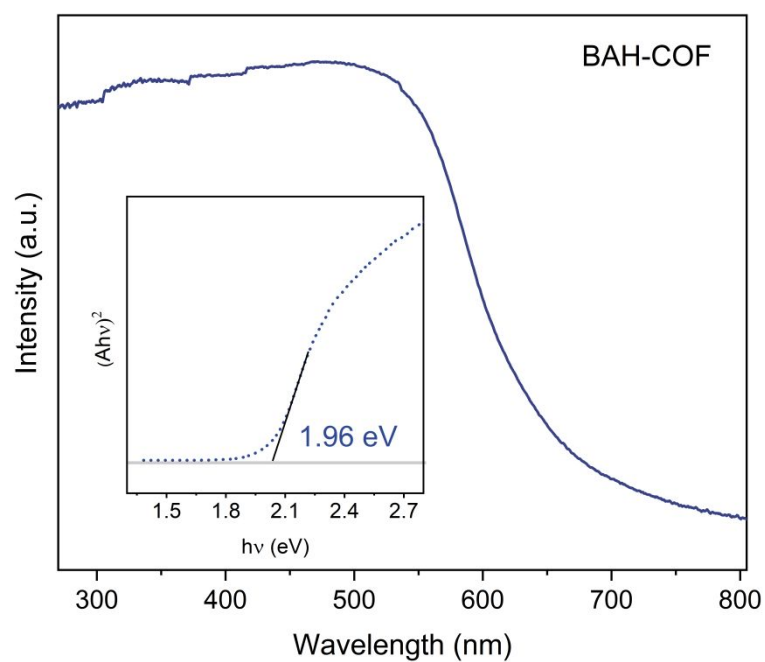


Figure S30. UV/vis diffuse reflectance spectra (DRS) spectra of BAH-COF. The insert figure is the corresponding Tauc plot.

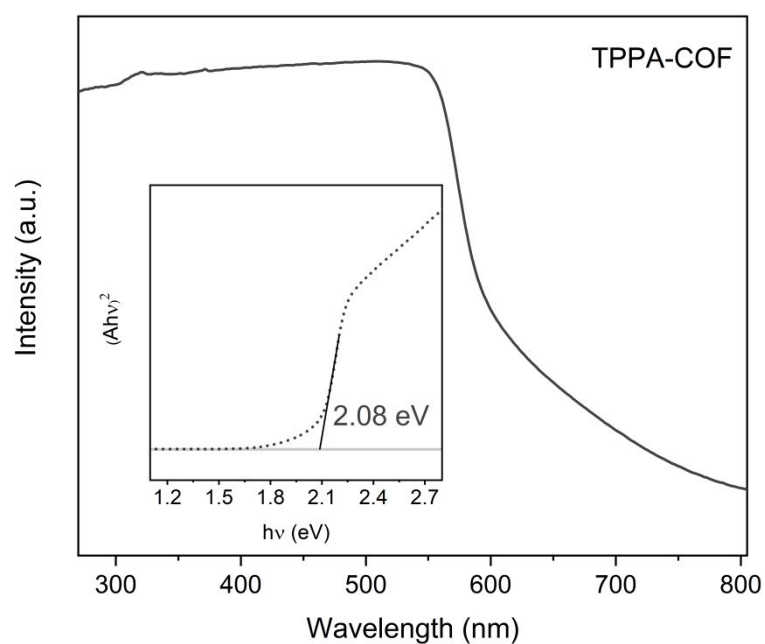


Figure S31. UV/vis diffuse reflectance spectra (DRS) spectra of TPPA-COF. The insert figure is the corresponding Tauc plot.

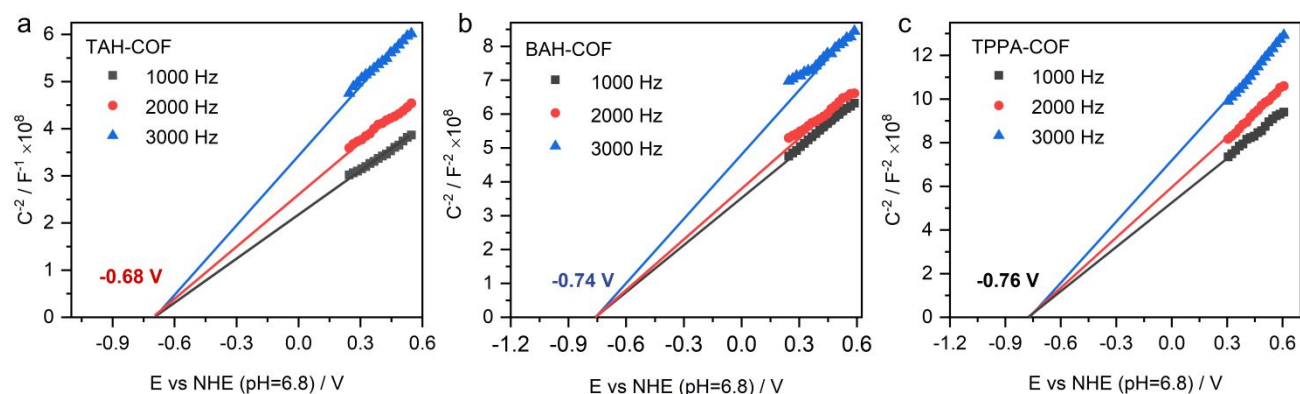


Figure S32. Mott-Schottky (M-S) plots for (a) TAH-COF, (b) BAH-COF and (c) TPPA-COF measured in 0.2 M Na₂SO₄ (pH = 6.8) with Ag/AgCl as the reference electrode in dark. (These vacuum level values were converted to electrochemical potentials according to: -4.5 eV versus vacuum level is equal to 0.0 V versus NHE at pH = 6.8).

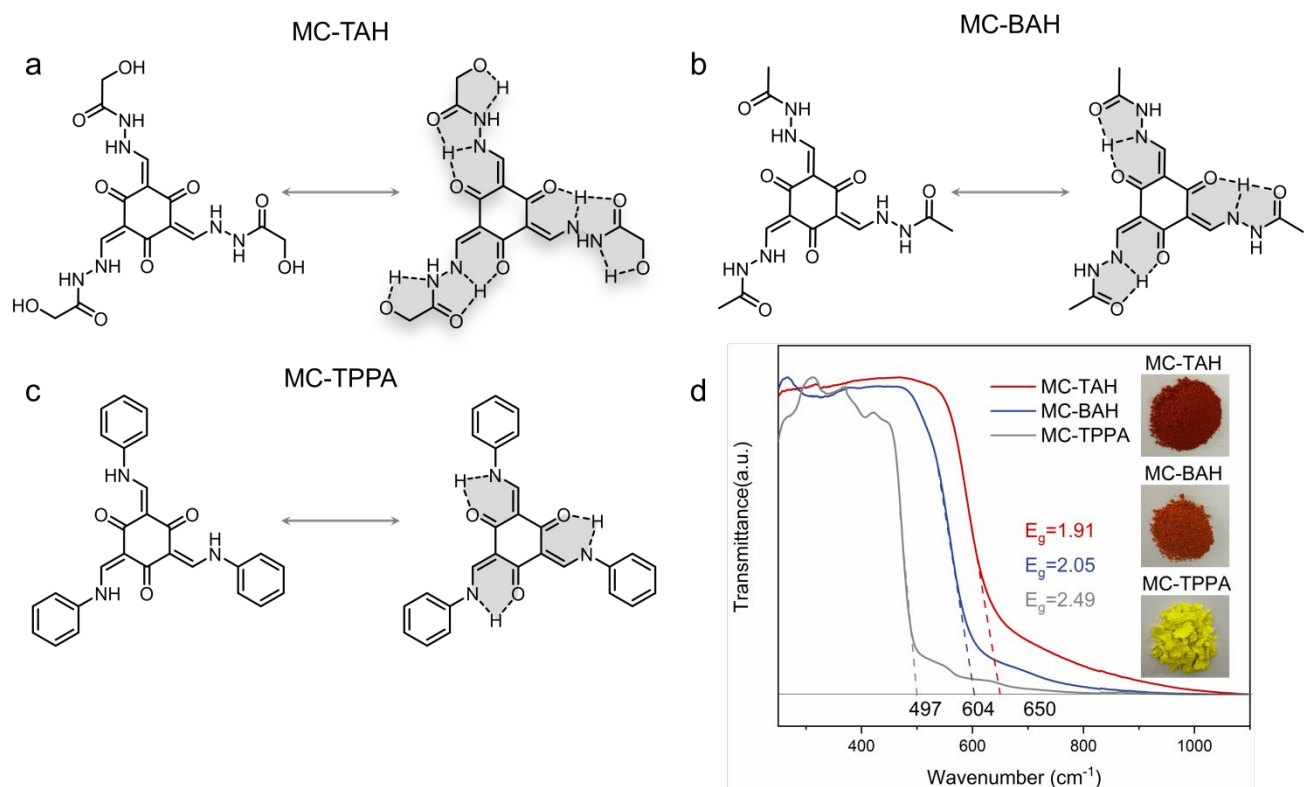


Figure S33. The RAHBs in tautomeric model compounds: (a) MC-TAH; (b) MC-BAH; (c) MC-TPPA. (d) The UV-vis absorption spectra and corresponding optical gaps for three model compounds. Inset shows the visual photographs of the compounds under ambient light.

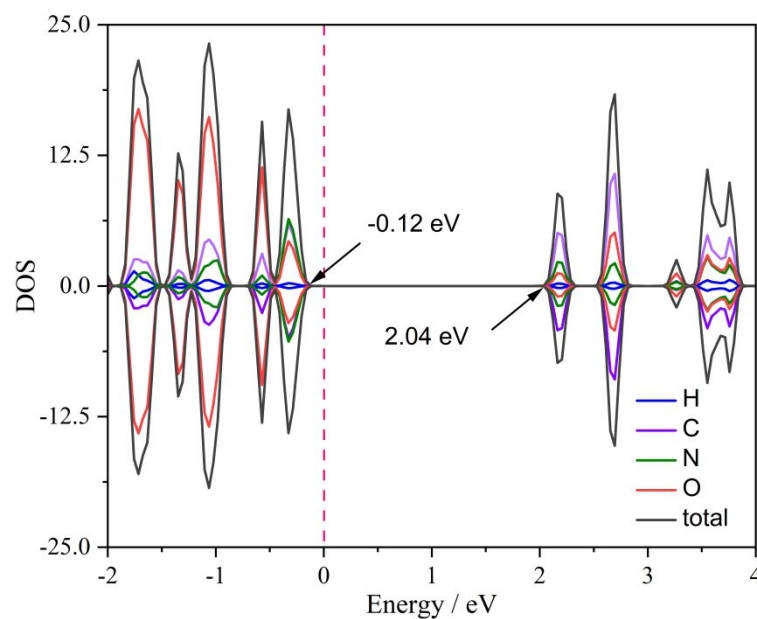


Figure S34. Electronic properties of TAH-COF using DFT calculations and corresponding density of states (DOS). The calculated band gap of 2.16 eV is in agreement with experimental measurements.

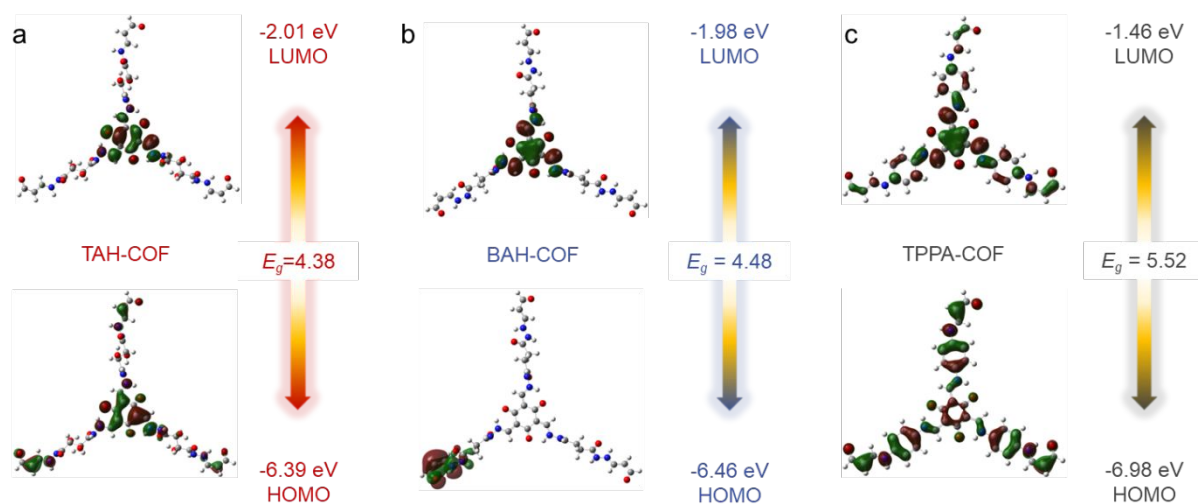


Figure S35. Highest occupied molecular orbital (HOMO) and Lowest unoccupied molecular orbital (LUMO) for the segments of (a) TAH-COF, (b) BAH-COF and (c) TPPA-COF. Noted that the estimated band gaps using DFT calculations are higher than the experimental optical band gaps, as the smallest repeating units without stacking were considered.

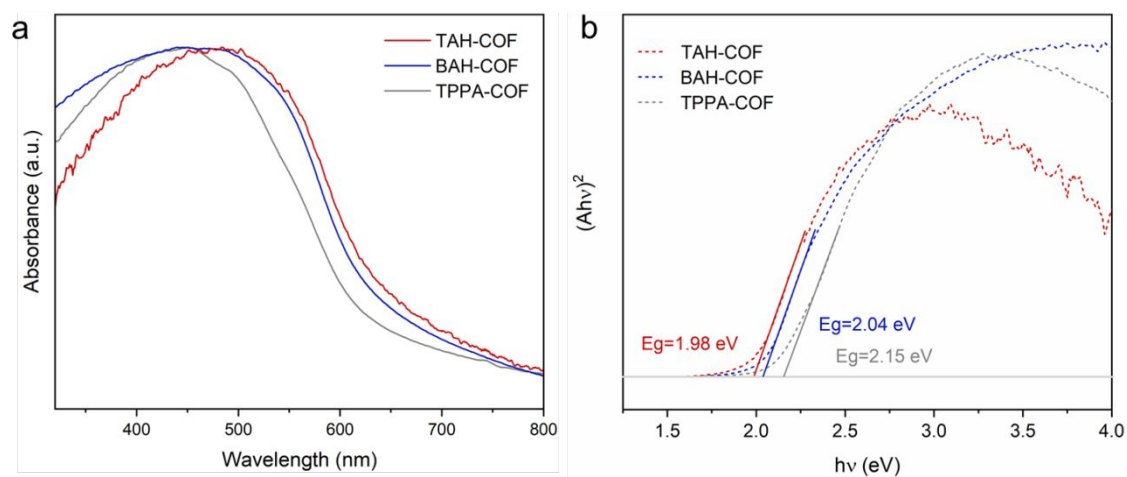


Figure S36. (a) The UV-Vis spectra of TAH-COF, BAH-COF, and TPPA-COF suspension in ethanol and (b) the corresponding Tauc plots. The band gaps were slightly wider than that obtained from diffuse reflectance spectra.

Table S1. Hydrogen bonding networks in per unit cell of TAH-COF, BAH-COF and TPPA-COF. The TAH-COF, which contains more polar groups in its main chain, exhibited a greater extent of hydrogen bonding.

	TAH-COF	BAH-COF	TPPA-COF
Hydrogen Bonds Min (number)	77	46	6
Hydrogen Bonds Max (number)	85	57	9
Average Hydrogen Bonds (number)	81	50	7
Min Length (Å)	1.76	1.83	1.67
Max Length (Å)	3.04	3.09	3.04
Average Length (Å)	2.30	2.37	2.16
Min Angle (°)	90.40	90.00	94.85
Max Angle (°)	179.44	161.33	149.42
Average Angle (°)	125.23	114.06	123.94

Section 4. Photocatalytic performance

4.1 Photocatalytic H₂O₂ evolution

Reactions were carried out in a Pyrex top-irradiation reaction vessel. Firstly, 5 mg of COF powder was added to 50 mL of deionized water. It was further ultrasonicated for 30 min to obtain a well-dispersed suspension. The mixture was magnetically stirred at a stirring speed of 800 rpm under visible-light illumination (300 W Xe-lamp, $\lambda > 420$ nm) in unsealed conditions. The temperature of the reaction solution was maintained at 25 °C by the flow of cooling water. The reaction mixture was sampled at specific time intervals. Upon completion of the reaction, the amount of H₂O₂ was analyzed by well-established iodimetry method^{S3}. The amount of I₃⁻ was determined by means of UV-vis spectroscopy on the basis of the absorbance at 350 nm, from which the amount of H₂O₂ produced during each reaction was estimated. After the photocatalysis experiment, COFs were recovered by washing with MeOH and THF before drying at 120 °C.

4.2 AQY measurements

The apparent quantum yield (AQY) was measured under the illumination of a 300 W Xe lamp with different band-pass filters ($\lambda 0 \pm 20$ nm) for 1 h. The photocatalyst (5 mg) was added to pure deionized water (50 mL). Ultrasound and O₂ bubbling for half an hour respectively, and the apparent quantum yields were calculated using the following equations:

$$\text{AQY}(\%) = \frac{2 \times \text{Number of evolved H}_2 \text{ molecules}}{\text{Number of incident photons}} \times 100\% = \frac{2 \times C \times N_A}{S \times P \times t \times \frac{\lambda}{h \times c}} \times 100\%$$

Where C is the H₂ production amount (μmol) per hour; N_A is the Avogadro constant ($6.02 \times 10^{23} \text{ mol}^{-1}$); S is the irradiation area (20.8 cm^2); P is the monochromatic light intensity (W cm^{-2}) (P is detected by optical power meter); t is the light irradiation time (3600 s); λ is the wavelength of the monochromatic light (nm); h is the Plank constant ($6.626 \times 10^{-34} \text{ J s}$); c is the speed of light ($3 \times 10^8 \text{ m s}^{-1}$).

4.3 SCC efficiency measurements

The solar-to-chemical energy conversion (SCC) efficiency was determined by using an AM 1.5 G solar simulator as the light source (300 W Xe lamp). The photocatalytic reaction was carried out in pure deionized water (50 mL) with photocatalysts (20 mg). Ultrasonication and O₂ bubbling for half an hour respectively. During the photocatalytic tests, O₂ was continually bubbled into the bottle. The SCC efficiency was calculated via the following equation:

$$\text{SCC efficiency}(\%) = \frac{\Delta G_{\text{H}_2\text{O}_2} \times n_{\text{H}_2\text{O}_2}}{t_{\text{ir}} \times S_{\text{ir}} \times I_{\text{AM}}} \times 100\%$$

Where $\Delta G_{\text{H}_2\text{O}_2} = 117 \text{ kJ mol}^{-1}$ is the free energy for H₂O₂ generation (117 kJ mol^{-1}). $n_{\text{H}_2\text{O}_2}$ is the amount of H₂O₂ generated and t_{ir} is the irradiation time (s). The overall irradiation intensity (I_{AM}) of the AM1.5 global spectrum (300–2,500 nm) is $1,000 \text{ W m}^{-2}$ and the irradiation areas (S_{ir}) are $4 \times 10^{-4} \text{ m}^2$. For example, when using TAH-COF as the catalyst, $81.0 \mu\text{mol H}_2\text{O}_2$ was generated during 1 h of illumination, and the SCC efficiency can be calculated as 0.66%.

4.4 Photocatalytic H₂ evolution

The photocatalytic H₂ generation experiments were conducted in a glass gas-closed-circulation system (CEL-SPH2N) under irradiation with a 300 W xenon lamp (CEL-HXF 300). A 5 mg COF photocatalyst was added to a 50 mL 0.1 M aqueous ascorbic acid solution, along with a 3% wt platinum (Pt) as cocatalyst. Hexachloroplatinic acid was used as the Pt precursor. Before light irradiation, the dissolved air must be thoroughly removed by a vacuum pump. And then the aqueous suspension was irradiated from the top using a xenon lamp. The temperature of the reaction solution was maintained at 25 °C by the flow of cooling water. The reaction mixture was kept under constant stirring using a magnetic stirring bar, and the evolved gases were analyzed by online gas chromatography (Shimadzu, GC-8A), with a thermal conductive detector and 5 Å molecular sieve column, using Ar as the carrier gas.

4.5 Photocatalytic O₂ evaluation

The photocatalytic O₂ generation experiments were carried out in a glass gas-closed-circulation system (CEL-SPH2N) under irradiation with a 300 W xenon lamp (CEL-HXF 300). The 5 mg photocatalyst was added to 50 mL of deionized water containing 100 mg of AgNO₃ and 100 mg of La₂O₃. The dissolved air must be thoroughly removed with a vacuum pump prior to irradiation using an AM 1.5G solar simulator as the light source (300 W Xe lamp). The temperature of the reaction solution was maintained at 25 °C by the flow of cooling water. The amount of O₂ was analyzed using an online gas chromatograph (GC 2060) with a thermal conductive detector and 5 Å molecular sieve column, using Ar as the carrier gas.

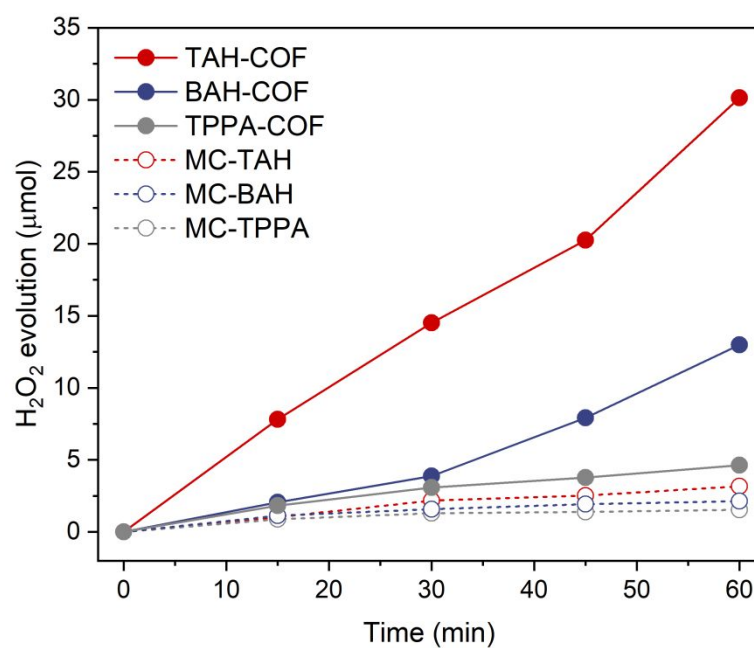


Figure S37. Time course of H_2O_2 evolution for three COFs and their corresponding model compounds (AM 1.5G irradiation, 50 mL pure water, 5 mg COF catalyst).

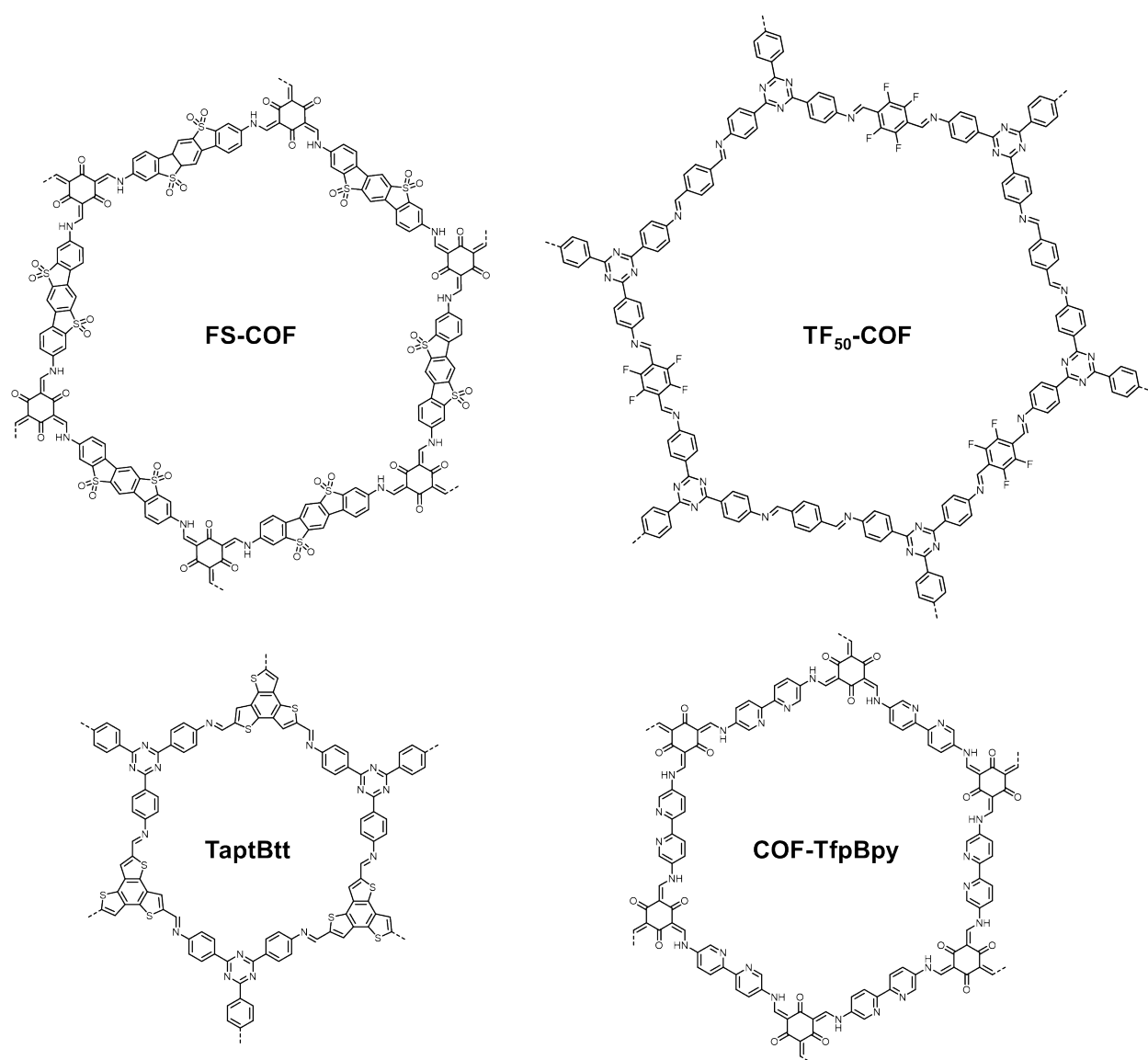


Figure S38. Schematic representation of state-of-the-art COFs for photocatalytic H₂O₂ evolution.

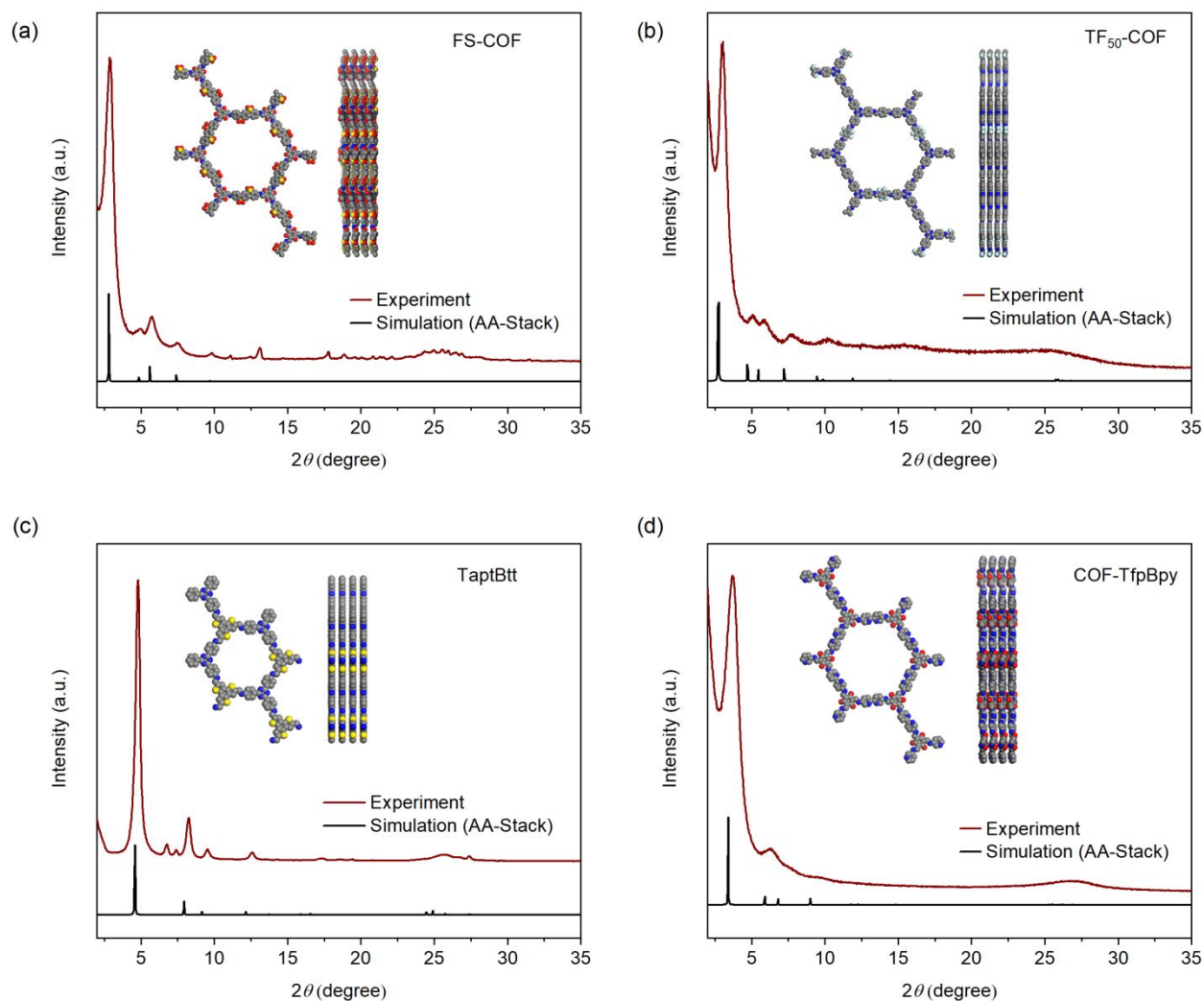


Figure S39. PXRD profiles of (a) FS-COF; (b) TF₅₀-COF; (c) TaptBtt; (d) COF-TfpBpy of the experimentally observed (red), simulated AA stacking modes (blue).

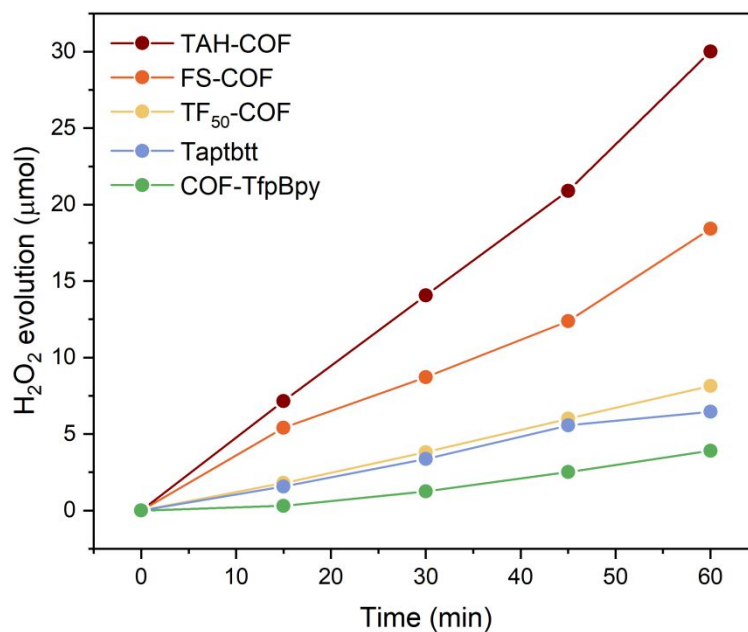


Figure S40. Time courses of photocatalytic H₂O₂ production for TAH-COF, FS-COF, TF₅₀-COF, TaptBtt and COF-TfpBpy.

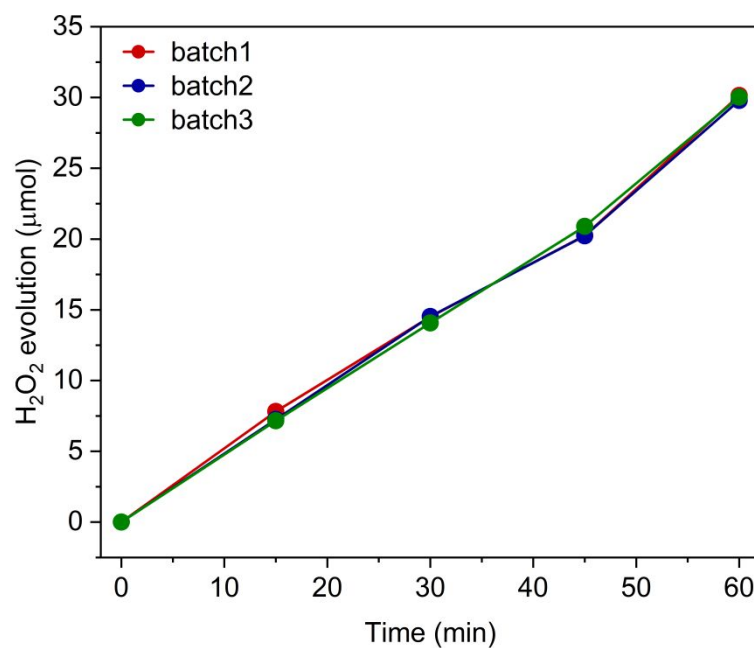


Figure S41. Time-dependent H₂O₂ evolution for TAH-COF from three different synthetic batches.

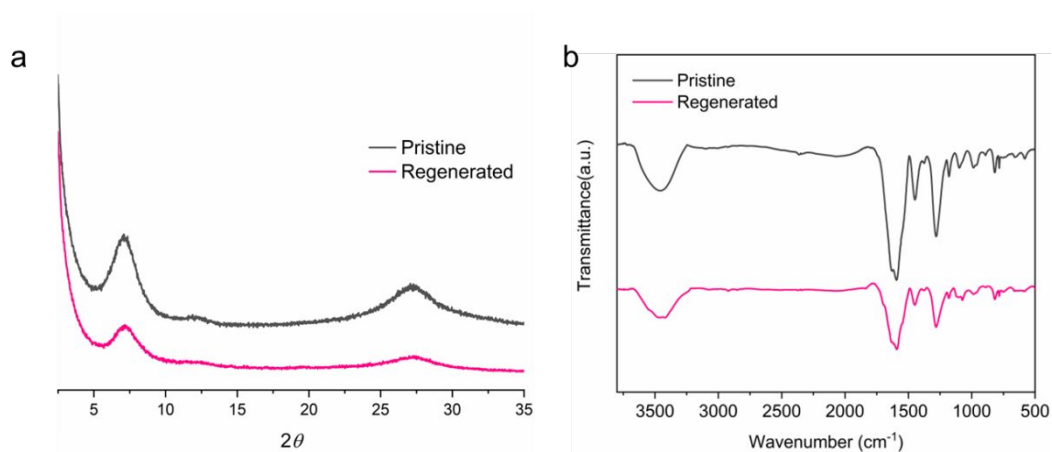


Figure S42. The PXRD patterns and FT-IR spectrum of TAH-COF before and after photocatalysis, which showed retention of crystalline structure and demonstrated the high stability of TAH-COF as a photocatalyst.

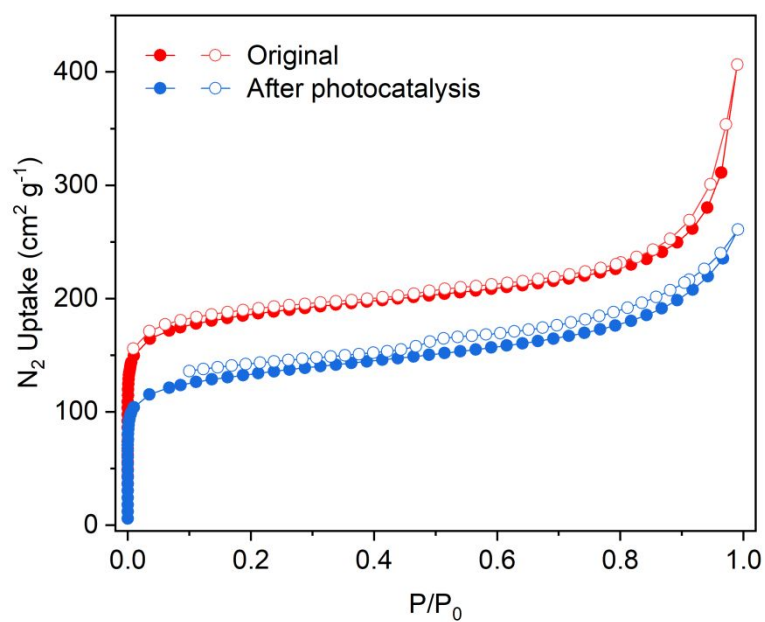


Figure S43. N_2 adsorption-desorption isotherms for TAH-COF before and after photocatalysis test. The calculated BET surface areas are 703, and 498 $m^2 g^{-1}$ for TAH-COF before and after the photocatalysis test, respectively.

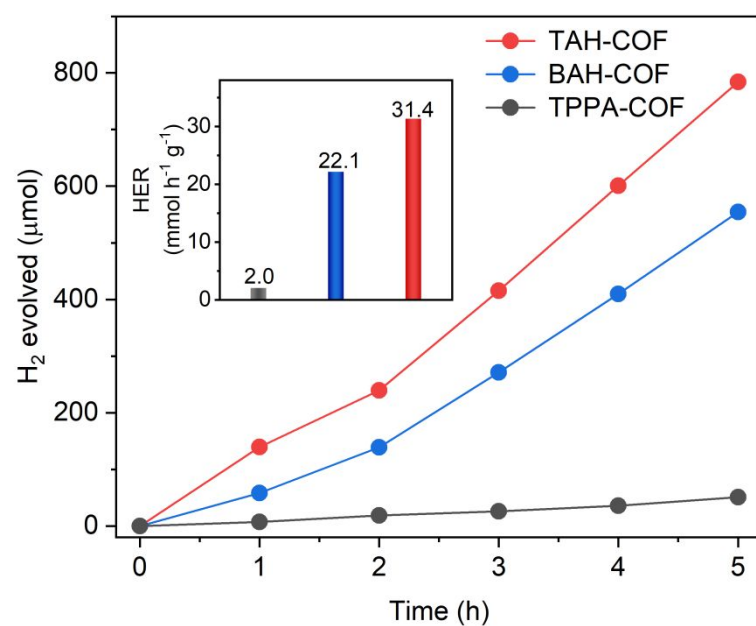


Figure S44. Time courses of sacrificial photocatalytic hydrogen production for TAH-COF, BAH-COF, and TPPA-COF.

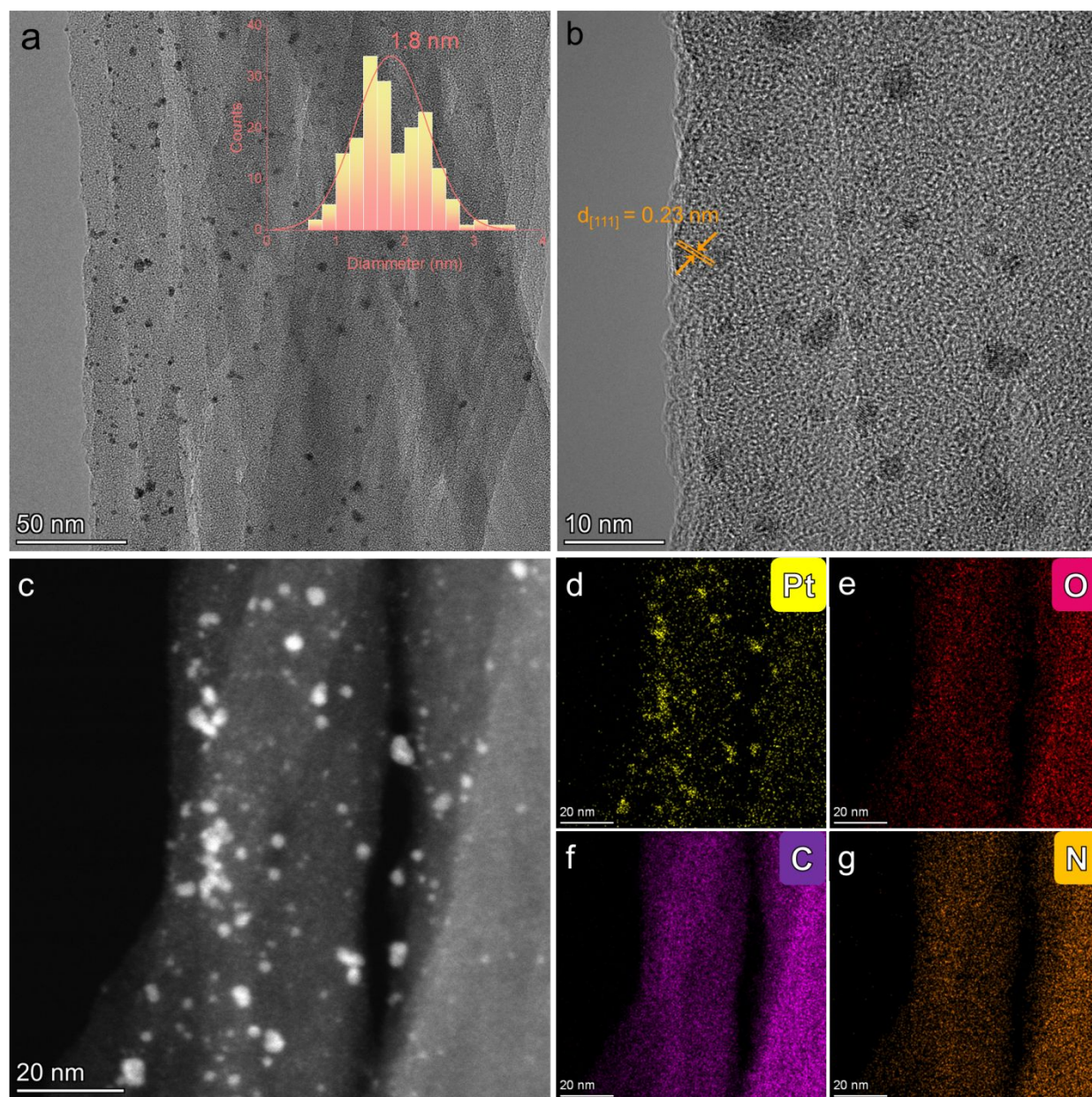


Figure S45. (a) HRTEM image for Pt nanoparticles on TAH-COF. Inset image is size distribution of Pt nanoparticles. (b) Enlarged HRTEM image shows [111] planes of Pt with d -spacing of 0.23 nm. (c) HAADF-STEM and (d) elemental mapping images of Pt doped TAH-COF.

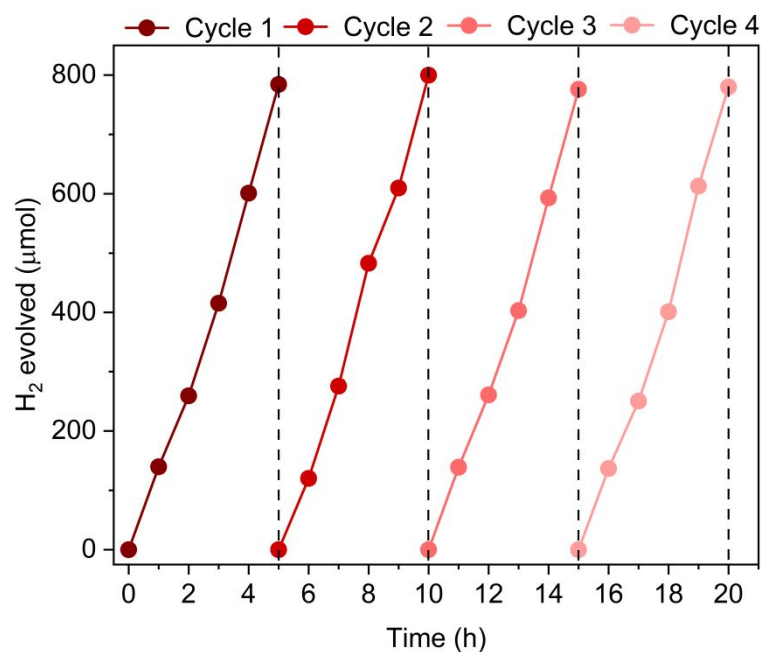


Figure S46. Long-term cycling experiments of TAH-COF for photocatalytic production of H₂.

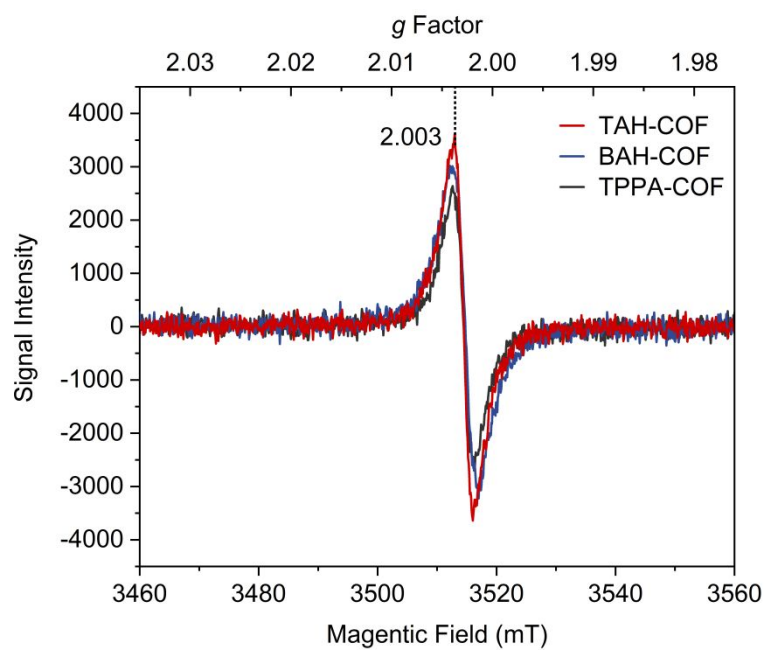


Figure S47. Electron spin resonance (ESR) of TAH-COF (red), BAH-COF (blue) and TPPA-COF (dark) upon light irradiation.

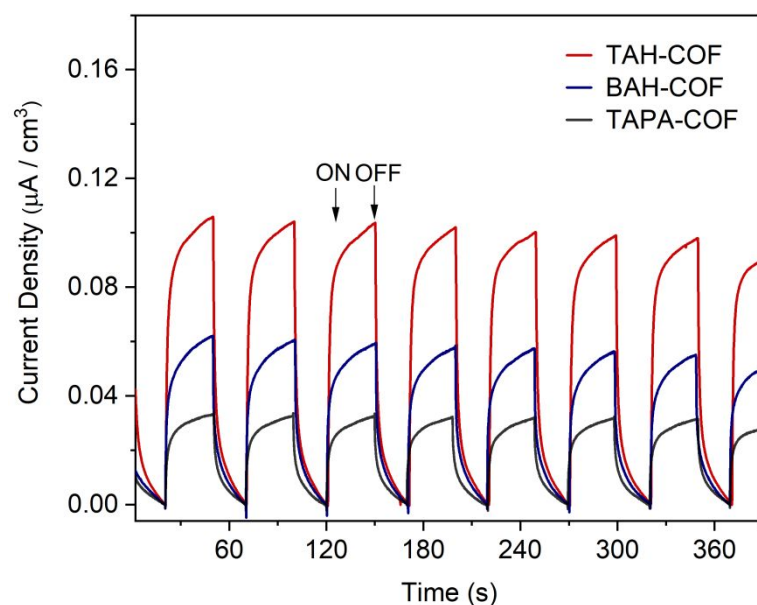


Figure S48. Transient photocurrents of the TAH-COF, BAH-COF, and TPPA-COF under visible light irradiation.

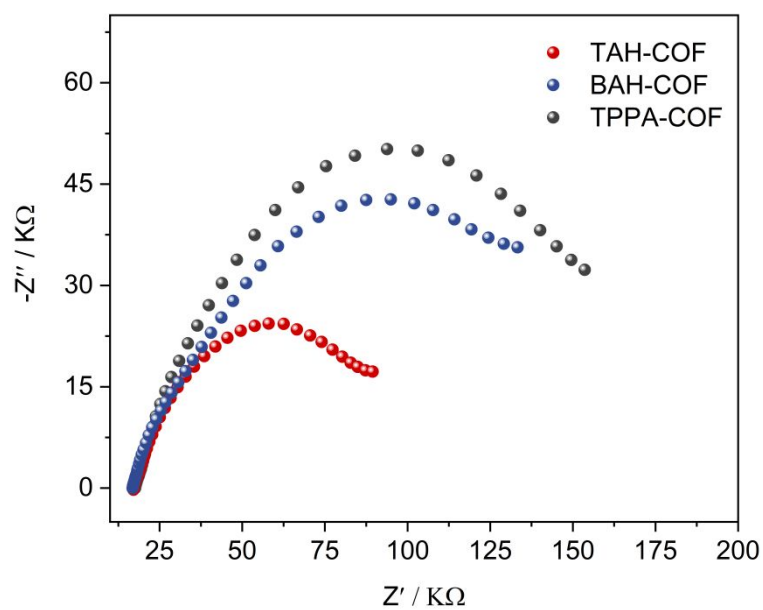


Figure S49. Electrochemical impedance spectra (EIS) spectra of TAH-COF, BAH-COF, and TPPA-COF.

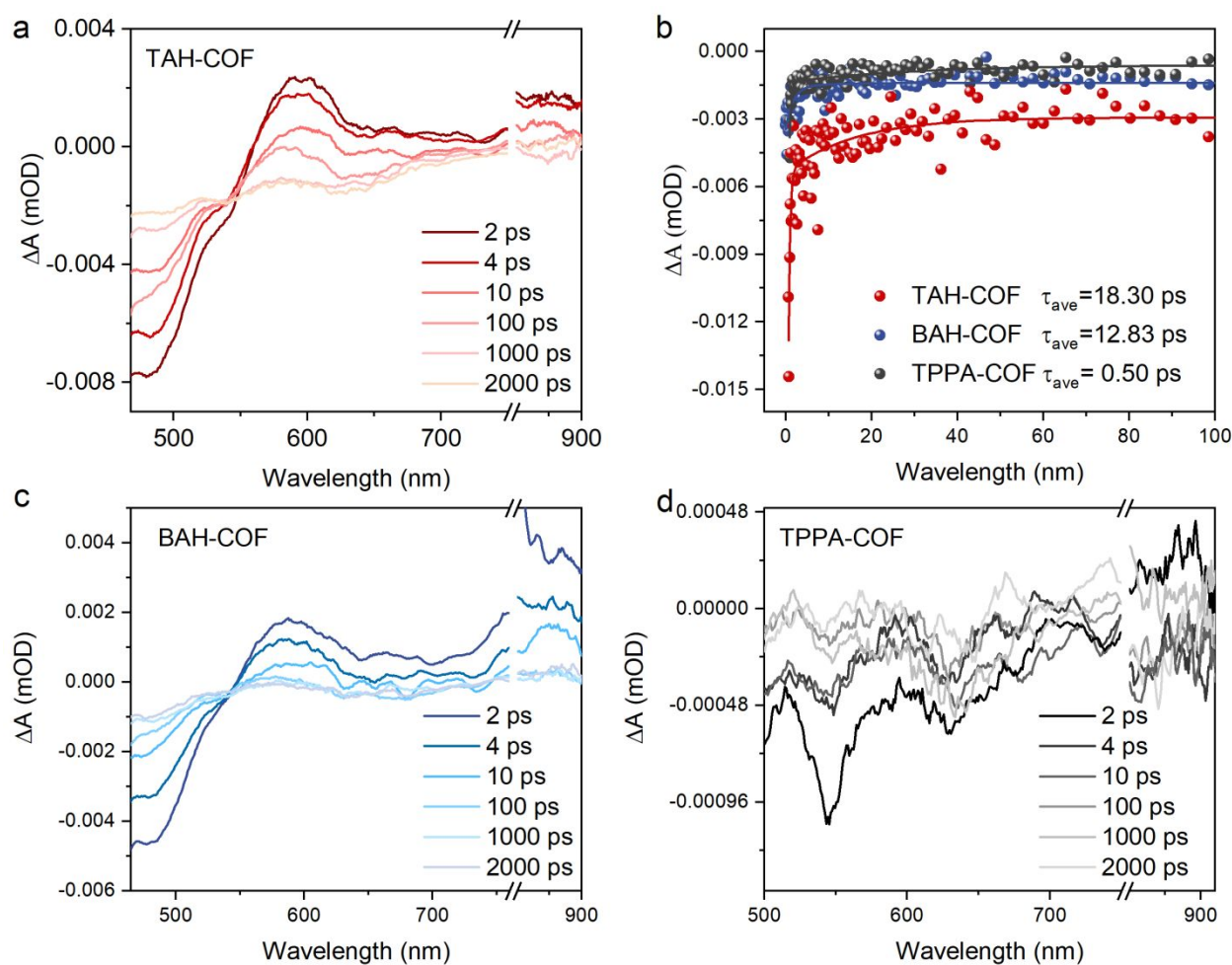


Figure S50. (a) Transient absorption spectra (TAS) of TAH-COF suspension in ethanol were performed upon 410 nm laser excitation. Negative signals were observed at <550 nm, which were attributed to the ground state bleach (GSB) associated with the band-edge, while positive signals were detected at >550 nm, corresponding to photoinduced excited-state absorption (PIA). (b) Kinetic trace plots of TAH-COF, BAH-COF at 485 nm and TPPA-COF at 550 nm. Transient absorption spectra (TAS) of (c) BAH-COF and (d) TPPA-COF suspension in ethanol were performed upon 410 nm laser excitation.

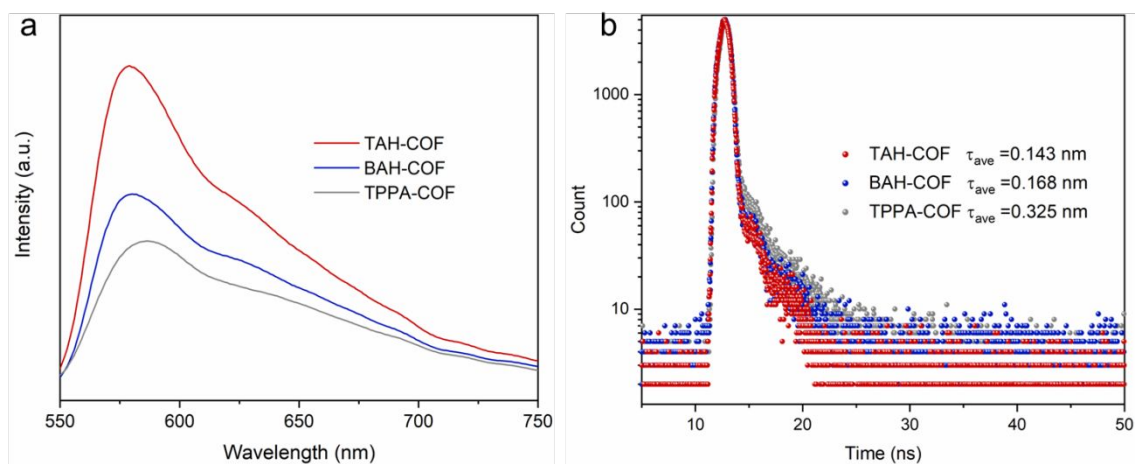


Figure S51. (a) Room temperature (298 K) steady-state photoluminescence (PL) spectrum for TAH-COF, BAH-COF, and TPPA-COF excited at 470 nm, (b) Time-resolved photoluminescence (TRPL) spectra for TAH-COF, BAH-COF, and TPPA-COF in ethanol (excitation wavelength at 470 nm and detected wavelength at 575 nm).

Table S2. Comparison of H₂O₂ production performance for different COF photocatalysts reported in the literatures.

Photocatalysts	Solvent	Illumination (nm)	H ₂ O ₂ ($\mu\text{mol h}^{-1} \text{g}^{-1}$)	SCC / AQY	Ref.
TAH-COF	H ₂ O	> 420	6003	SCC=0.66 % AQY=7.72 % (500 nm)	This work
CTF-BDDBN	H ₂ O	> 420	97.2	SCC=0.14%	S4
TAPD-(Me) ₂ COF	H ₂ O:EtOH=9:1	420-700	97	/	S5
SonoCOF-F2	H ₂ O	>420	164	AQY=4.8% (420 nm)	S6
	Water:BA=1:9	>420	415		
HEP-TAPT-COF	H ₂ O	>420	1750	SCC=0.65% AQY=15.35% (420 nm)	S7
HEP-TAPB-COF	H ₂ O	>420	990	SCC=0.38% AQY=9.98% (420 nm)	
Py-Py-COF	H ₂ O: BA = 9:1	> 420	1242	SCC= 0.09% AQY= 4.5 % (420 nm)	S8
COF-TfpBpy	H ₂ O	>420	694.7	SCC=0.57%; AQY=8.1% (420 nm)	S9
		>300	1313	SCC=1.08%; AQY=13.5% (420 nm)	
DETH-COF	H ₂ O	>420	1665	/	S10
TF ₅₀ -COF	H ₂ O:EtOH=9:1	> 400	1739	SCC=0.17 % AQY=5.1 % (400 nm)	S11
EBA-COF	H ₂ O : EtOH = 9:1	=420	1830	AQY=4.4% (400 nm)	S12
CoPc-BTM-COF	H ₂ O : EtOH = 9:1	>400	2096	AQY=8.6% (630 nm)	S13
CoPc-DAB-COF	H ₂ O : EtOH = 9:1	>400	1851	AQY=5.2% (630 nm)	
DE7-M	H ₂ O	>420	2216	SCC=0.28%; AQY=8.7% (420 nm)	S14
TTF-BT-COF	H ₂ O	>420	2760	SCC=0.49% AQY=11.19% (420 nm)	S15
AQTEE-COP	H ₂ O	≥400	3204	AQY=3.59% (400 nm)	S16
Bpt-CTF	H ₂ O	350-780	3268.1	SCC=0.2% AQY = 8.6% (400 nm)	S17
FS-COFs	H ₂ O	>400	1506.1	AQY = 6.21% (420 nm)	S18
		>420	3904		
Bpy-TAPT	H ₂ O	>420	4038	SCC=0.65 % AQY=8.6 % (420 nm)	S19

Note: / = not described

Table S3. Comparison of H₂ production performance for different COF photocatalysts reported in the literatures.

COF	Co-catalyst	Sacrificial donor	Illumination	HER ($\mu\text{mol g}^{-1} \text{h}^{-1}$)	Ref
TAH-COF	Pt (3 wt %)	0.1 M AA	AM 1.5G	31400	This work
sp ² c-COF _{ERDN}	Pt (3 wt %)	TEOA (10 vol.%)	>420 nm	2120	S20
TFPT-COF	Pt (2.2 wt %)	TEOA (10 vol.%)	>420 nm	1970	S21
N ₃ -COF	Pt (0.68 wt %)	TEOA (10 vol.%)	>420 nm	1703	S22
COF-alkene	Pt (3 wt %)	TEOA (20 vol.%)	>420 nm	2330	S23
DCNA-1 AC	8 wt% H ₂ PtCl ₆	0.1 M AC	>420 nm	27886.67	S24
Tp-2C/BPy ²⁺ -COF	Pt (3 wt %)	0.1 M AC	>420 nm	34600	S25
CYANO-COF	Pt (1 wt %)	0.1 M AA	>420 nm	60850	S26
CYANO-CON	Pt (1 wt %)	0.1 M AA	>420 nm	134200	
TP-BDDA COF	Pt (3 wt %)	TEOA (10 vol.%)	>395 nm	324	S27
NKCOF-108	Pt (5 wt %)	0.1 M AA	>420 nm	11600	S28
Pt-PVP-TP-COF	Pt (6 wt %)	0.05M AA	>420 nm	8420	S29
BtCOF150	Pt (1 wt %)	TEOA (20 vol.%)	>400 nm	750	S30
PY-DHBD-COF	Pt	0.01M AA	>420 nm	42432 (1 wt% Pt) 71160 (3 wt% Pt)	S31
Py-CITP-BT-COF	none	0.1 M AA	>420 nm	8875	S32
30%PEG@BT-COF	Pt (3.5 wt %)	0.1 M AA	>420 nm	11140	S33
TtaTfa	8 wt% H ₂ PtCl ₆	0.1 M AA	>420 nm	20700	S34
ZnPor-DETH-COF	8 wt% H ₂ PtCl ₆	TEOA (10 vol.%)	>400 nm	413	S35
RC-COF-1	Pt (3 wt %)	0.1 M AA	>420 nm	27980	S36
g-C ₁₈ N ₃ -COF	Pt (3 wt %)	1M AA	>420 nm	292	S37
N2-COF	chloro(pyridine) cobaloxime	TEOA (10 vol.%)	AM 1.5G	782	S38
FS-COF	Pt (3 wt %)	0.1 M AA	>420 nm	10100	S39
TpDTz	nickelthiolate hexameric	TEOA (10 vol.%)	AM 1.5G	941	S40
A-TEBPY-COF	Pt (2.2 wt %)	TEOA (10 vol.%)	AM 1.5G	98	S41
TPPA(Δ)-Cu(II)-COF	none	0.1M L-Cysteine	>420 nm	14700	S42
PyTz-COF	Pt (3 wt %)	0.1 M AA	AM 1.5G	2072.4	S43
USTB-10	Pt (3 wt %)	0.1 M AA	>420 nm	21800	S44
g-C ₄₀ N ₃ -COF	Pt (3 wt %)	TEOA (10 vol.%)	>420 nm	4120	S45
g-C ₅₄ N ₆ -COF	Pt (3 wt %)	TEOA (10 vol.%)	>420 nm	2518.9	S46

Section 5. Mechanism of photocatalytic H₂O₂ production

5.1 Rotating disk electrode measurements

A rotating disk electrode served as the substrate for the working electrode. 2 mg of COFs were dispersed in a mixture containing 180 μL isopropanol and 20 μL Nafion solution (5 wt%) and ultrasonic treated to form homogenous catalyst ink. The mixture (10 μL) was put onto the surface of a glassy carbon disk (diameter: 5 mm; geometric area: 0.196 cm^2) and dried at room temperature. RDE measurements were carried out on the CHI 760D workstation (CH Instruments, Inc.) with an RRDE-3A rotator (ALS Co., Ltd). A Graphite rod and Ag/AgCl electrodes were used as counter electrodes and reference electrodes, respectively. The linear sweep voltammograms (LSV) were obtained in an O₂-saturated 0.1 M phosphate buffer solution (PBS, pH = 7.2) solution at room temperature with a scan rate of 10 mV s^{-1} and different rotation speeds. The average number of electrons (n) was calculated by Koutecky-Levich equation:

$$\frac{1}{J} = \frac{1}{J_K} + \frac{1}{J_L} = \frac{1}{J_K} + \frac{1}{B \omega^{1/2}}$$

$$B = 0.2nFC_{O_2}D_{O_2}^{2/3}\nu^{-1/6}$$

where J is the current density, J_K and J_L are the kinetic and diffusion-limiting current densities, ω is the rotating speed (rpm), n is the transferred electron number, F is Faraday constant (96485 C mol^{-1}), C_{O_2} is the bulk concentration of O₂ ($1.26 \times 10^{-3} \text{ mol cm}^{-3}$), D_{O_2} is the diffusion coefficient of O₂ ($2.7 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$), and ν is kinetic viscosity of the electrolyte ($0.01 \text{ cm}^2 \text{ s}^{-1}$), respectively.

5.2 In-situ FT-IR spectrometry measurements

An FT-IR spectrometer (PerkinElmer Frontier) equipped with an in situ diffuse reflection cell (Harrick Praying Mantis) was used for in situ DRIFTS analysis. About 20 mg of the COFs was placed into the sample holder. The catalyst was activated under the N₂ atmosphere for 6 h and cooled down to 25 $^{\circ}\text{C}$ in N₂. Afterward, the O₂ and H₂O vapors were introduced into the cell and fiber source. The spectra were collected every 10 minutes until the reaction is in equilibrium.

5.3 Electrochemical measurement

The Mott-Schottky analysis, electrochemical impedance spectra (EIS), and photocurrent responses ($I-t$) of samples were also obtained on a CHI650E electrochemical workstation in a standard three-electrode configuration (Pt wire as counter electrode, and Ag/AgCl as reference electrode). The working electrode preparation process is as follows: Mix 10 mg of samples and 5 μL of a Nafion (5 wt% Nafion in isopropanol) aqueous solution with ethanol as solvent and grind into a slurry. The working electrodes were prepared by spreading 50 μL homogeneous suspension over the surface of ITO glass substrate ($1.5 \text{ cm} \times 1.5 \text{ cm}$) and then filmed at 50 $^{\circ}\text{C}$ in a vacuum oven. For Mott-Schottky experiments, the partitioned signal was 10 mV with a frequency of 1000, 2000, and 3000 Hz. The $I-t$ measurements were performed with the light on and off in a 0.5M Na₂SO₄ aqueous solution, and the bias was 0 V versus Ag/AgCl. The EIS measurements were performed at the open-circuit voltage with an amplitude of 5 mV in a frequency range of 10² Hz to 10⁶ Hz. A

Xe lamp (300W)fitted with cut-off filters (≥ 420 nm) was used as the light source for the EIS measurement. The applied potentials versus Ag/AgCl are converted to NHE potentials using the following equations:

$$E_{NHE} = E_{Ag/AgCl} + E^0_{Ag/AgCl} (E^0_{Ag/AgCl} = 0.199 \text{ V})$$

5.4 Computational details

All geometric optimization and total energy calculations in this study were performed at the spin-polarized DFT by using the Vienna Ab initio Simulation Package (VASP)^{S47-S50}. The projector augmented wave (PAW) method^{S51, S52} and the Perdew-Burke-Ernzerh (PBE)^{S53} functional under the generalized gradient approximation (GGA)^{S54} were applied throughout the calculations, and a cut-off energy of 400 eV was used. To avoid interactions between neighboring slabs, a vacuum layer of ~ 15 Å was set. We constructed altogether three COF surfaces (BAD-COF, TAD-COF and TpPa-COF), and each surface model is kept relaxed. The Brillouin zones were sampled with k-points of $2 \times 2 \times 1$ Gamma grids for different COF systems calculations. The convergence criteria of energy and force in the calculations were set as 1.0×10^{-4} eV and 0.05 eV/Å, respectively.

Under acidic conditions, the total reaction for O₂ reduction to H₂O₂ is:



The ORR process occurring on the different COF surfaces follows the 2 electron association mechanism that involves two elementary steps:



where * and *OOH represent an empty adsorption site on the surface and the adsorbed OOH species, respectively. Firstly, the computational hydrogen electrode (CHE) model⁵ was used to obtain the reaction free energy of the ($\text{H}^+ + \text{e}^-$) transfer step, *i.e.*, the free energy of ($\text{H}^+ + \text{e}^-$) was equal to the chemical potential of $1/2$ H₂ under the actual reaction conditions ($T = 298$ K, $P = 1$ bar). Correspondingly, the Gibbs free energy change ΔG of each elementary step in the ORR process was calculated as follows:

$$\Delta G = \Delta E + \Delta \text{ZPE} - T \times \Delta S \quad (\text{S4})$$

where ΔE is the reaction energy calculated by DFT and the changes of zero-point energy (ΔZPE) and entropy contribution ($T \times \Delta S$) were obtained by calculating the frequency of the system, and we set the reaction temperature as 298.15 K. The ZPE was calculated as follows⁹:

$$\text{ZPE} = \sum h\nu / 2000 \quad (\text{S5})$$

The partition function and entropy were calculated as follows^{S55}:

$$x_i = \frac{100ch\nu_i}{k_B T} \quad (\text{S6})$$

$$pf_i = \frac{x_i}{e^{x_i} - 1} - \log(1 - e^{-x_i}) \quad (\text{S7})$$

$$S = \frac{\sum (R \times pf_i)}{1000 \times 96.485} \quad (\text{S8})$$

where c , h , ν , k_B , T and R are the light speed ($\text{m} \times \text{s}^{-1}$), Plank constant, frequency (cm^{-1}), Boltzman constant, temperature (298.15 K) and gas constant, respectively. The units of ZPE and S are eV and eV/K, respectively. i is the number of vibration frequencies, and each adsorbed molecule has

3n vibration frequencies (n is the number of atoms). In addition, the frequencies of small gas phase molecules (H_2 , O_2 and H_2O) were calculated using the Gaussian 16 program package^{S56} and 6-311G* basis sets^{S57} were used to obtain their ZPE and S values.

Since it is difficult to obtain the free energies of OOH species in electrolyte solutions, the free energies of adsorption of G_{OOH^*} are relative to those of $H_2O(g)$ and $H_2(g)$ in the gas phase defined as follows:

$$G_{O_2^*} = G_{tot} - G^* - 2G_{H_2O} + 2G_{H_2} \quad (S9)$$

$$G_{OOH^*} = G_{tot} - G^* - 2G_{H_2O} + 3/2G_{H_2} \quad (S10)$$

where G_{tot} represents the overall free energy after adsorption of the intermediate, and G^* represents the free energy of clean surface. Obviously, the negative value of G_{X^*} indicates the exothermic adsorption process. The more negative the G_{X^*} is, the stronger the binding between adsorbed OOH* species and the surface is.

In addition, the O_2 , H_2O and OOH adsorption energy ($E_{ads}(O_2)$, $E_{ads}(H_2O)$ and $E_{ads}(OOH)$) can be calculated as follows:

$$E_{ads}(O_2) = -(E_{O_2^*} - E^* - 2E_{H_2O} + 2E_{H_2}) \quad (S11)$$

$$E_{ads}(OOH) = -(E_{OOH^*} - E^* - (2E_{H_2O} - 3/2E_{H_2})) \quad (S12)$$

$$E_{ads}(H_2O) = -(E_{H_2O^*} - E^* - E_{H_2O}) \quad (S13)$$

where $E_{O_2^*}$ and E_{OOH^*} are the total energies of the different COF systems after O_2 and OOH adsorption, respectively, and E^* is the total energy of the different COF systems. E_{O_2} , E_{H_2O} and E_{H_2} are the calculated total energies of O_2 , H_2O and H_2 molecules in the vacuum, respectively. Larger (more positive) E_{ads} indicates the stronger adsorption.

To get insight into the electronic properties of BAD-COF, TAD-COF and TpPa-COF, the Gaussian 16 program package^{S56} was used for molecular frontier orbitals calculations at the M06-2X level^{S58}, the TZVP basis set^{S59} for all atoms. To calculate the solvent effect (water), the self-consistent reaction field method based on the SMD model^{S60} was utilized.

Ab-initio molecular dynamics simulations. One used the Ab-initio molecular dynamics (AIMD) simulations were performed with the Vienna ab-initio Simulation Package (VASP)^{S47-S50}. The Perdew–Burke–Ernzerhof^{S53} parametrization of generalized gradient approximation was adopted to describe the exchange and correlation term. The TAD-COF contains 48 water molecules and the TpPa-COF contains 17 water molecules. The cutoff energy of the plane-wave basis was set to 400 eV. Only the gamma point was used to sample the Brillouin zone. At the start of the AIMD simulation, the COF samples in the supercell were assigned an initial temperature of 300 K. The time step was chosen to be 1.0 fs, and the integration of Newton's equation was based on the Verlet algorithm implemented in VASP. The AIMD simulation for investigating the solvation structure of the COF was then performed for 20 ps.

5.5 The equation of $2e^-$ two-steps and h^+ redox process

The process of $2e^-$ ORR (Eq. 1-3) and $4h^+$ WOR (Eq. 4) proceeds based on the following equations:





where * represents the surface-active site.

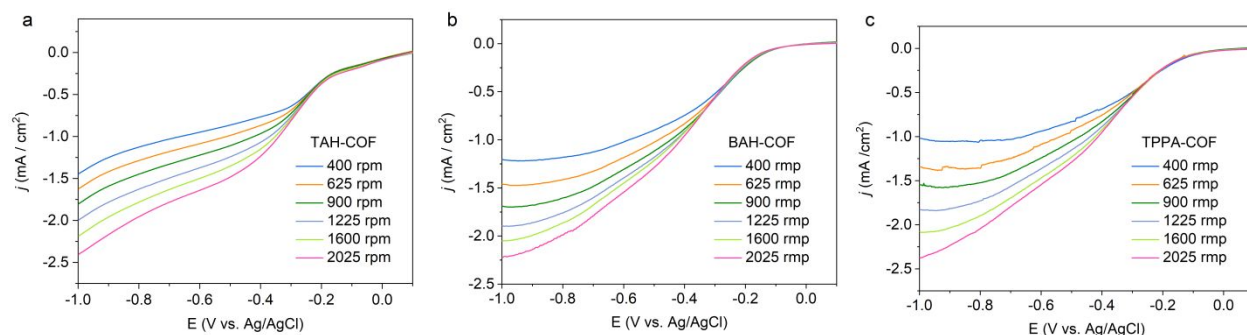


Figure S52. Linear sweep voltammetry (LSV) curves of (a) TAH-COF, (b) BAH-COF and (c) TPPA-COF recorded on a rotating disk glassy carbon electrode in 0.1 M phosphate buffer solution (PBS, pH = 7.2) solution saturated with O_2 in dark condition.

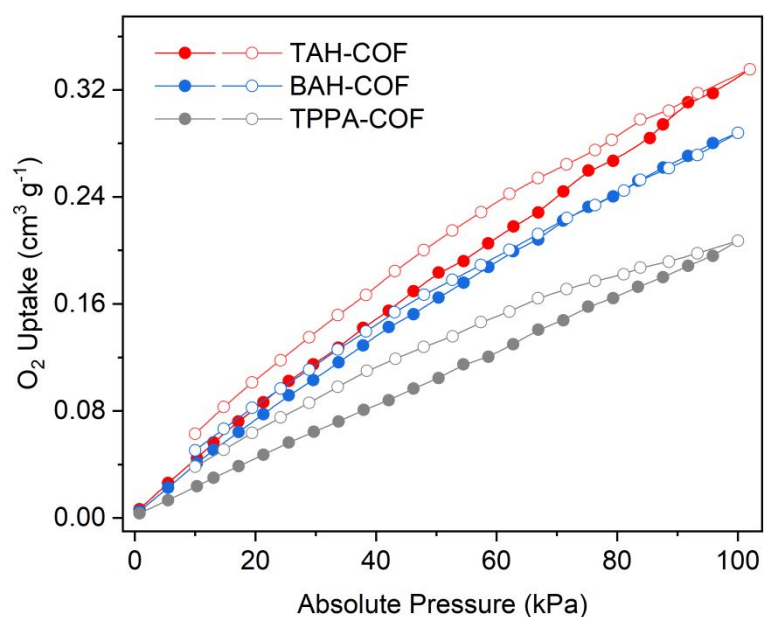
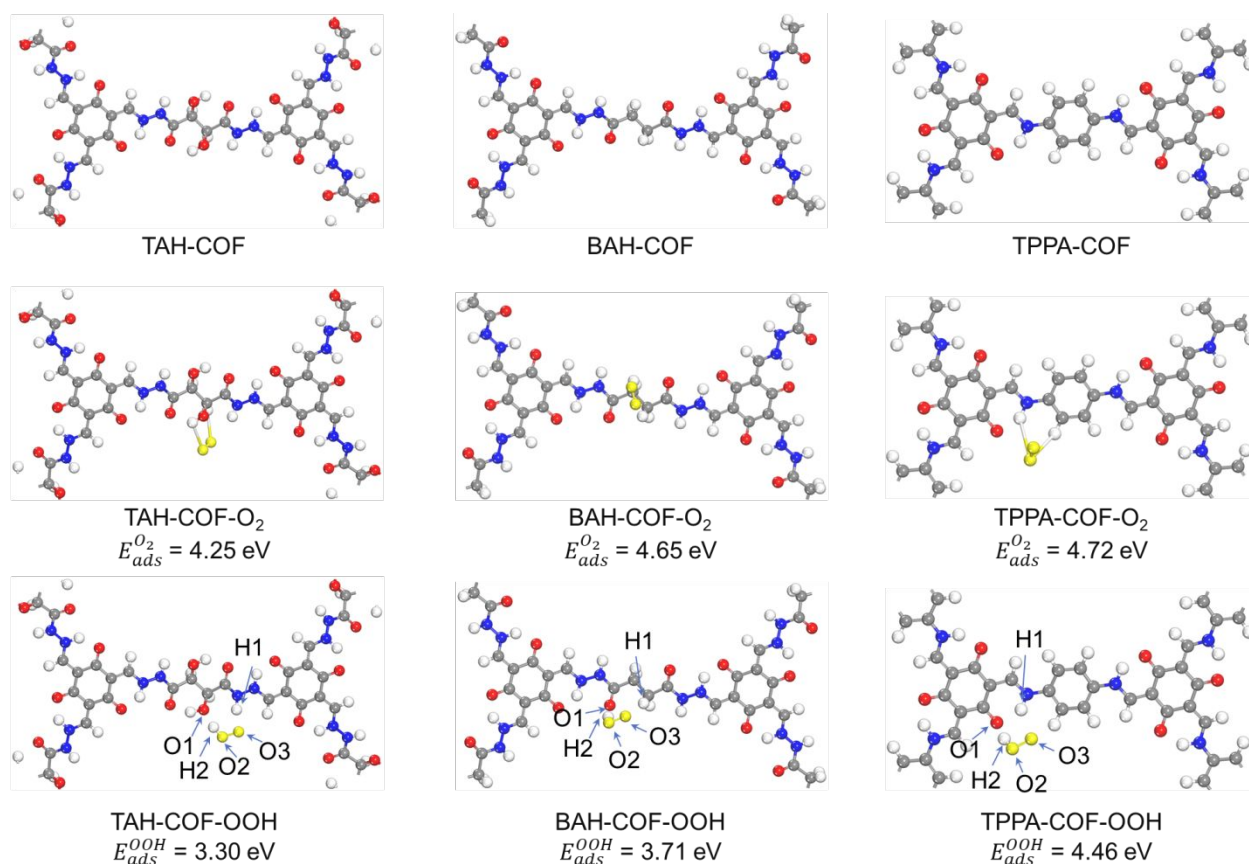


Figure S53. Adsorption (filled circles) and desorption (open circles) isotherms for O_2 uptake in TAH-COF, BAH-COF and TPPA-COF recorded at 273 K.



Sites	Bader Charge / e				
	O1	H1	O2	O3	H2
TAH-COF	-1.159	0.580	-0.488	-0.240	0.679
BAH-COF	-1.152	0.143	-0.528	-0.195	0.696
TPPA-COF	-1.120	0.105	-0.471	-0.224	0.643

Figure S54. Calculated the Bader charges of OOH species and key adsorption sites on TAH-COF, BAH-COF and TPPA-COF. It was observed that TAH-COF exhibited strong adsorption of O₂ and OOH* with E_{ads} values of 4.25 eV and 3.30 eV, lower than that of BAH-COF and TPPA-COF. Moreover, the Bader charge for H1 site in TAH-COF is the most positive, which contributes to the enhanced stability of the O species in the OOH species. Note that the *OOH intermediate forms two hydrogen bonds with the hydroxyl and amide groups of TAH-COF. All optimized structures are presented in a top-view perspective. The white, grey, blue, and red spheres in framework refer to hydrogen, carbon, nitrogen and oxygen atoms, respectively. The yellow and white in O₂ and OOH* refer to oxygen and hydrogen atoms.

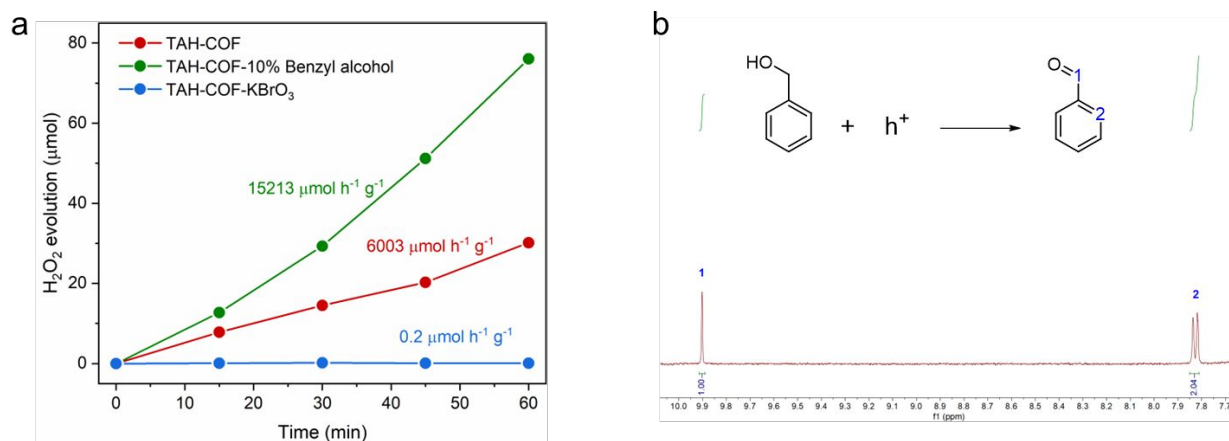


Figure S55. (a) Comparison of H_2O_2 evolution by TAH-COF under electron and hole scavengers (10% BA and 10 mM KBrO_3). (b) ^1H NMR spectra of the end product after 1 h illumination reaction by TAH-COF.

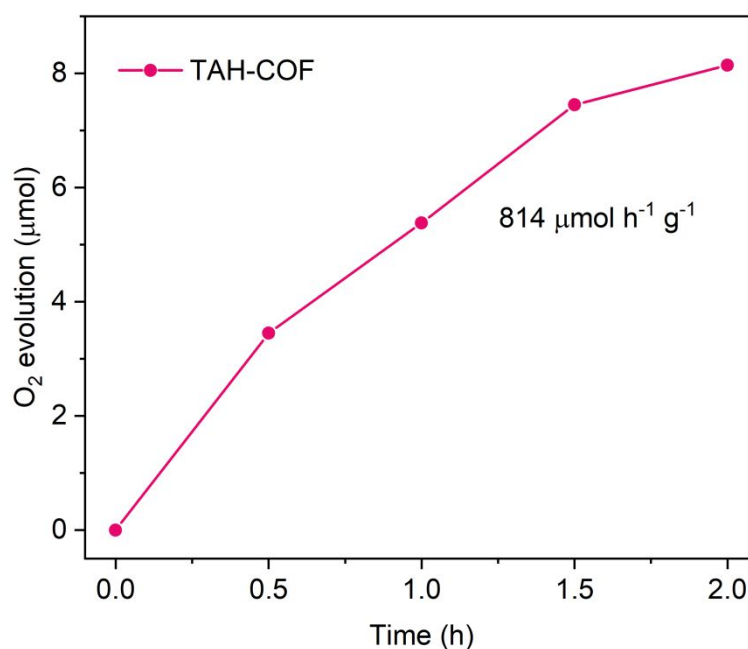


Figure S56. Photocatalytic O_2 evolution performance of TAH-COF under AM 1.5G irradiation using NaIO_3 as the electron sacrificial agent.

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